Robust Optimal Agile Flight of Quadrotors: An Internal Model-Based Nonlinear Model Predictive Optimization Approach

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***Abstract*—Achieving robust agile quadrotor flight under unknown external disturbances and parametric uncertainty remains a significant challenge. Inspired by robust output regulation theory, this paper proposes $\mathcal{IM}$-NMPO, an $\mathcal{I}$nternal $\mathcal{M}$odel-based Nonlinear Model Predictive Optimization framework for robust optimal planning and control during agile flight. The core insight is the systematic decoupling of uncertain dynamics from the optimization loop via the internal model principle (IMP), enabling real-time disturbance learning and rejection while reducing reliance on high-fidelity physical models. This allows both time-optimal planning and nonlinear model predictive control (NMPC) to operate on the same disturbance-decoupled nominal dynamics. The proposed framework comprises three main components: translational and rotational nonlinear internal model compensators for real-time disturbance rejection, an NMPC optimization stabilizer for constrained agile trajectory tracking, and a receding-horizon polynomial waypoint allocation module for efficient online time-optimal reference generation. Robust constraint satisfaction, recursive feasibility and asymptotic stability are guaranteed through rigorous theoretical analysis. The effectiveness and generalizability are validated through extensive flight experiments under various disturbances, including unknown payloads, persistent fan-induced winds, and time-varying gusts, across quadrotors with different wheelbases. The framework was successfully deployed in the 2024 DJI Robomaster Intelligent MAV Championship for Planning and Control of Quadrotors, where it completed challenging racing courses at speeds up to 23.3 m/s, ranking first and finishing in less than one-third of the time taken by the second-place.**

***Index Terms*—Internal model, nonlinear model predictive optimization, disturbance rejection, agile quadrotor flight.**



Fig. 1: Time-lapse illustrations of agile quadrotor flight racing through narrow gates under unknown external disturbances and parametric uncertainty. Video recordings are supplied in Movie 1[1].

I. Introduction

A. Motivation

SAFETY and agility subject to unknown external disturbances and parametric uncertainty are of great significance for micro aerial vehicles (MAVs). Search and rescue missions require MAVs to visit multiple waypoints within limited time, while aerial delivery requires precise navigation through constrained spaces with varying payloads and unknown winds [1]–[3]. Although existing optimization-based methods have demonstrated considerable advantages in agile flight, they typically rely on precise models for accurate prediction, which are often unattainable due to varying wind conditions, payload variations, and unmodeled aerodynamic effects [4], [5]. These limitations underscore the necessity to improve disturbance rejection and environmental adaptability in agile flight.

Nonlinear model predictive control has achieved notable success in aggressive flight, with speeds exceeding 22 m/s [6], [7]. However, such performance usually depends on high-fidelity aerodynamic models [8], whose accuracy degrades under unmodeled disturbances. Data-driven methods [1], [9] help mitigate tracking errors, but usually require prior environmental data and large-scale datasets for effective training.

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[1]https://www.youtube.com/

Robust NMPC extends standard NMPC by providing formal guarantees under disturbances and model uncertainty [10]–[12], but its computational burden and tube-design complexity limit real-time deployment in agile flight.

Recent methods [13]–[15] have applied output regulation theory [16], [17] to address external disturbances, effectively solving the trajectory tracking and/or disturbance rejection problems as regulation problems. Nevertheless, these approaches generally rely on feedback control and typically do not explicitly account for system constraints, thereby lacking the ability to anticipate future reference information. These limitations pose significant challenges for achieving accurate agile flight under external disturbances, while operating near the motion limits of the quadrotor.

Many MAV missions are time-critical. Given limited battery capacity, time-efficient flight is essential for improving task efficiency [18]. However, state-of-the-art time-optimal planners [6], [19] struggle with large-scale and dynamic scenarios. Moreover, model uncertainty arising from inaccurate aerodynamic models and physical parameters, often leads to suboptimal planning and control performance. These challenges have motivated benchmark competitions such as the Autonomous Drone Race in IEEE IROS, the AlphaPilot Challenge, and the DJI Robomaster Intelligent MAV Championship [20].

B. Contributions

This paper addresses the fundamental challenges of robust agile quadrotor flight under unknown external disturbances and parametric uncertainty, aiming to enhance the **agility** of quadrotors while maintaining **safety** and **robustness** (Fig. 1 and Movie 1). The overall framework is illustrated in Fig. 2, and the primary contributions are outlined as follows:

1) $\mathcal{IM}$**-NMPO framework**: We propose an $\mathcal{I}$nternal $\mathcal{M}$odel-based Nonlinear Model Predictive Optimization framework that systematically decouples uncertain dynamics from the optimization loop via the internal model principle, actively learning and rejecting disturbances in real-time. This eliminates the need for defining a tube or adjusting its size through local feedback controllers as in robust NMPC, enabling robust optimal planning and control while reducing reliance on high-fidelity models.
2) **Theoretical properties**: Rigorous theoretical properties, including robust constraint satisfaction, recursive feasibility, and asymptotic stability, are guaranteed through a hierarchical framework: i) translational and rotational nonlinear internal model compensators for real-time disturbance rejection; ii) a nominal system-based NMPC with tightened constraints to ensure recursive feasibility; and iii) an adaptive feedback controller to handle system parametric uncertainty and bound deviations between the original uncertain system and the nominal system.
3) **Novelty and generalizability**: To the best of our knowledge, this is the first integration of **IMP** with **NMPO** for robust agile quadrotor flight under disturbances and uncertainty. A receding-horizon polynomial-based waypoint allocation module is further introduced for efficient online time-optimal reference generation. The framework is validated across quadrotors with different wheelbases under various disturbances, including unknown payloads, persistent fan-induced winds, and unpredictable time-varying gusts. Its practical efficacy is demonstrated by winning the 2024 DJI Robomaster Intelligent MAV Championship. The source code and data have been released at https://github.com/Fly-to-the-Mars/IM_NMPO.git.

C. Organization

The remainder of this paper is organized as follows. Section II reviews related work and summarizes the remaining challenges. Section III introduces the preliminaries and problem formulation. Section IV presents the main results. In particular, a robust agile flight framework is established, including two classes of nonlinear internal model compensators, a polynomial-based time-optimal online planner, and a nominal system-based NMPC optimization stabilizer. Robust constraint satisfaction, recursive feasibility, and stability are also provided. Sections V and VI present simulation and real-world experiments. Finally, Section VII concludes the paper.

II. Related Work

A. Quadrotor Flight Control with Disturbances

Existing approaches for handling disturbances in agile quadrotor flight mainly focus on two aspects: rapid replanning and robust control. The former updates the trajectory online based on the current state [21], or inserts an adaptive preprocessing layer between the planner and controller [19]. However, the lack of explicit modeling for disturbances limits responsiveness to unpredictable or persistent external disturbances, potentially resulting in trajectory tracking degradation.

For robust tracking control, disturbance rejection remains a central challenge. Inspired by output regulation theory [16], [17], the IMP has been used to handle disturbances [13], [22]. A class of nonlinear internal models is introduced in [13] to achieve global asymptotic tracking when disturbances are composed of unknown sinusoidal signals. However, classical output-regulation-based feedback controllers lack the ability to anticipate future reference information and fail to handle system constraints, which potentially leads to constraint violations or closed-loop instability in agile flight [23].

Model predictive control (MPC) solves a finite-horizon optimal control problem and has proven effective in handling system constraints and predicting future system trajectories [23]–[26]. With increasing onboard computation, full-state NMPC has drawn more attention [27]. However, NMPC typically relies on accurate aerodynamic models, whose fidelity deteriorates under disturbances and uncertainty. To improve robustness, hybrid NMPC frameworks have been investigated, such as combining NMPC with incremental nonlinear dynamic inversion (INDI) [27]. While these methods empirically improve disturbance rejection, they often lack formal guarantees of recursive feasibility and stability [28]. Tube-based robust MPC provides rigorous robustness guarantees for systems with additive disturbances [10]; for instance, [11] combines tube-based robust MPC with neural-network-based uncertainty approximation to achieve real-time safe feedback control.

Recent work on constrained nonlinear output regulation with NMPC avoids solving classical regulator equations [29].

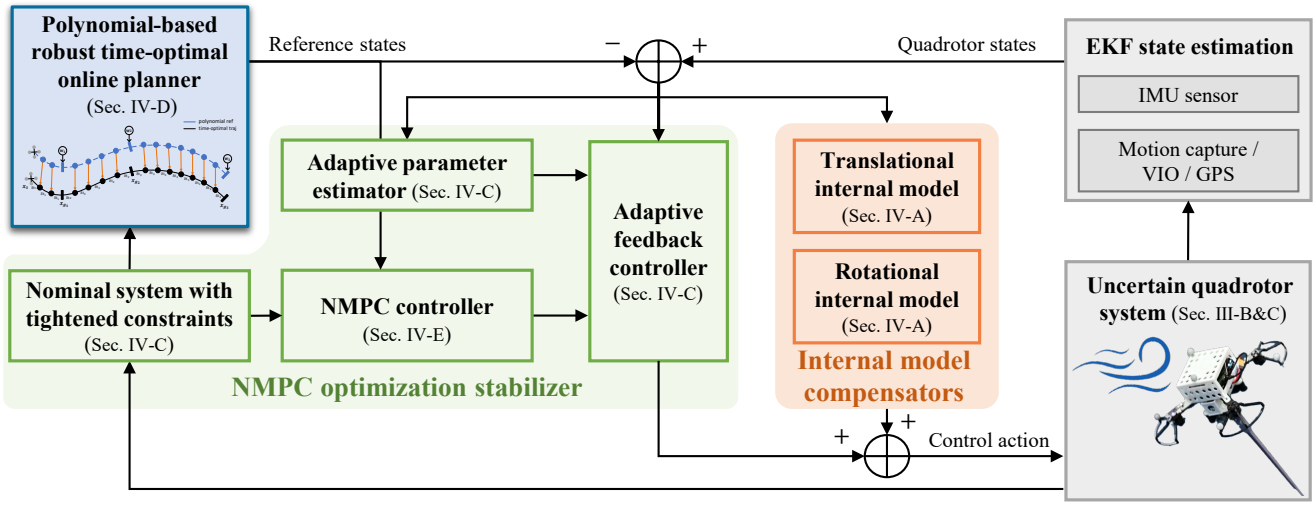


Fig. 2: $\mathcal{IM}$**-NMPO framework**: 1) two classes of nonlinear internal model compensators; 2) a polynomial-based robust time-optimal online planner; 3) a nominal system-based NMPC optimization stabilizer. This systematic design certifies end-to-end robust constraint satisfaction, recursive feasibility, and asymptotic stability for the original uncertain system.

The key limitation is the requirement of perfectly predictable disturbances, which is challenging to satisfy in real-world flight. These limitations motivate a framework that integrates online disturbance learning and rejection with constraint-aware predictive optimization, while reducing reliance on high-fidelity prior models or perfect disturbance predictability.

B. Time-Optimal Trajectory Planning for Quadrotors

Time-optimal planning with waypoint constraints is challenging due to the associated time allocation problem. Each waypoint constraint must be assigned to a specific time step, rendering the optimization difficult to solve. Existing methods can be broadly categorized into two paradigms: optimization-based and polynomial-based approaches.

Optimization-based methods explicitly address the time allocation problem within the optimization framework. A representative work [6] incorporates progress variables and complementary progress constraints (CPC) to generate truly time-optimal trajectories under full quadrotor dynamics. It has been extended to cluttered environments via a sampling-based framework [30]. While achieving optimality, their computational cost, often ranging from minutes to hours, limits real-time applicability. To improve efficiency, [19] manually specifies nodes with pre-assigned waypoint constraints and employs separate sampling periods between consecutive waypoints, while model predictive contouring control (MPCC) methods [21], [31] utilize an approximated point-mass model (PMM) for fast receding-horizon optimization. More recently, [32] develops a fast semidefinite convex relaxation for optimal control problems with spatio-temporal constraints, improving numerical tractability and computational efficiency.

Polynomial-based methods prioritize computational efficiency and trajectory smoothness by leveraging the differential flatness property. MINCO [33] provides an efficient piecewise-polynomial framework for handling multiple constraints, and [34] further uses MINCO for time-optimal gate-traversing trajectory optimization. Nevertheless, the inherent smoothness of polynomials generally results in time-suboptimal solutions.

While more sophisticated aerodynamic models exist [8], [35], their high computational cost limits adoption in real-time planning. Thus, the aforementioned time-optimal planning methods predominantly employ simplified linear-drag approximations [36] for computational tractability. These simplified models may become inaccurate during high-speed flight or under external disturbances [6]. Therefore, real-time agile flight requires planning methods that balance time-optimality, computational efficiency, and robustness to model uncertainty.

III. Preliminaries and Problem Formulation

A. Notation

Scalars are denoted by lowercase letters and vectors by bold lowercase letters. The world frame is represented as $\mathcal{W} : \{\boldsymbol{o}_W, \boldsymbol{x}_W, \boldsymbol{y}_W, \boldsymbol{z}_W\}$. The body frame is denoted by $\mathcal{B} : \{\boldsymbol{o}_B, \boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B\}$, with $\boldsymbol{z}_B$ aligning with the collective thrust direction, as depicted in Fig. 3. For $\boldsymbol{x}_i \in \mathbb{R}^{n_i \times m}, i = 1, \ldots, k$, $\text{col}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k) = \left[\boldsymbol{x}_1^\top, \ldots, \boldsymbol{x}_k^\top\right]^\top$. The orientation of $\mathcal{B}$ with respect to $\mathcal{W}$ is represented by the unit quaternion $\boldsymbol{q} = \text{col}[q_w, q_x, q_y, q_z] \in \mathbb{R}^4$ with $||\boldsymbol{q}|| = 1$, where $\boldsymbol{q}_v = \text{col}[q_x, q_y, q_z]$ and q_w are the vector and scalar parts, respectively. The n-dimensional identity matrix is denoted by $\boldsymbol{I}_n \in \mathbb{R}^{n \times n}$. For $\boldsymbol{p} \in \mathbb{R}^3$, $q(\boldsymbol{p}) = \text{col}(0, \boldsymbol{p})$. For a symmetric positive definite matrix $K \in \mathbb{R}^{n \times n}$, $\lambda_m(K)$ and $\lambda_M(K)$ represent the minimal and maximal eigenvalues of K, respectively. Operator block diag $(A_1, \ldots, A_n)$ denotes a block diagonal matrix with diagonal entries A_i. The symbol $\odot$ represents the Hamilton quaternion multiplication, while $\oplus$ and $\ominus$ denote Minkowski addition and subtraction, respectively. A set $\mathbb{X} \subseteq \mathbb{R}^n$ is a $\mathcal{C}$ set if it is compact, convex, and contains the origin in its interior.

B. Quadrotor Dynamics

The translational dynamics of the quadrotor are given by

$$\dot{\boldsymbol{p}} = \boldsymbol{v} \tag{1}$$

$$\dot{\boldsymbol{v}} = \boldsymbol{g} + f_T \boldsymbol{z}_B + m^{-1} \boldsymbol{d}_f \tag{2}$$

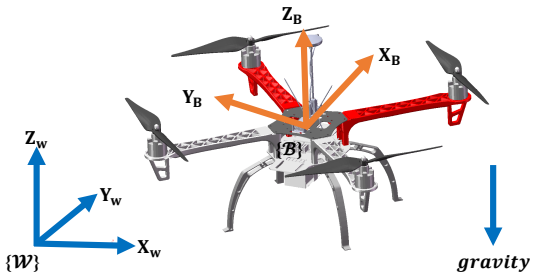


Fig. 3: Diagram of the quadrotor model with the world reference frame $\mathcal{W}$ and body-fixed frame $\mathcal{B}$.

where $\boldsymbol{p} = \text{col}\,[p_x, p_y, p_z] \in \mathbb{R}^3$ and $\boldsymbol{v} = \text{col}\,[v_x, v_y, v_z] \in \mathbb{R}^3$ denote the position and linear velocity of the quadrotor's center of gravity (CoG) with respect to $\mathcal{W}$, respectively; f_T and m are the mass-normalized collective thrust and total mass, respectively; $\boldsymbol{g} \in \mathbb{R}^3$ is the gravitational vector; $\boldsymbol{d}_f = \text{col}\,[d_1(t), d_2(t), d_3(t)] \in \mathbb{R}^3$ indicates all external disturbance forces acting on the quadrotor.

The rotational dynamics of the quadrotor are given by

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \odot q(\boldsymbol{\omega}) \tag{3}$$

$$J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega}^{\times} J\boldsymbol{\omega} + \boldsymbol{d}_\tau \tag{4}$$

where $\boldsymbol{q} = \text{col}\,[q_w, \boldsymbol{q}_v] \in \mathbb{R}^4$ is the unit quaternion representing the rotation from $\mathcal{W}$ to $\mathcal{B}$, with $q_w \in \mathbb{R}$ and $\boldsymbol{q}_v \in \mathbb{R}^3$; $\boldsymbol{\omega} = \text{col}\,[\omega_x, \omega_y, \omega_z] \in \mathbb{R}^3$ and $J \in \mathbb{R}^{3\times3}$ are the angular velocity of $\mathcal{B}$ with respect to $\mathcal{W}$ and the symmetric positive definite inertia matrix subject to parametric uncertainty, respectively. The uncertainty in J mainly arises from model-plant mismatch due to parameter identification errors or varying payloads. The resultant torque generated by rotors is denoted by $\boldsymbol{\tau} = \text{col}[\tau_x, \tau_y, \tau_z] \in \mathbb{R}^3$. The external disturbance torque is given by $\boldsymbol{d}_\tau = \text{col}[d_4(t), d_5(t), d_6(t)] \in \mathbb{R}^3$.

The quadrotor state $\boldsymbol{x}$ and control input $\boldsymbol{u}$ are defined as

$$\boldsymbol{x} = \text{col}\,[\boldsymbol{p}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}], \quad \boldsymbol{u} = \text{col}\,[f_T, \boldsymbol{\tau}] \tag{5}$$

which satisfy the physical constraints

$$\boldsymbol{x} \in \mathbb{X} \subseteq \mathbb{R}^n, \quad \boldsymbol{u} \in \mathbb{U} \subseteq \mathbb{R}^h \tag{6}$$

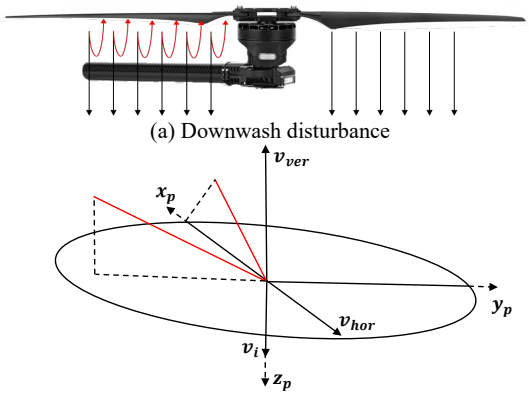
for some $\mathcal{C}$ sets $\mathbb{X}$ and $\mathbb{U}$, with $f_T \in \mathbb{U}_f$ and $\boldsymbol{\tau} \in \mathbb{U}_\tau$.

C. Model Uncertainty

We consider two sources of uncertainty: unknown external disturbances $\boldsymbol{d}(t) = \text{col}\,[\boldsymbol{d}_f, \boldsymbol{d}_\tau]$ and uncertain inertia parameters in J. The external disturbances mainly include unknown wind disturbances and propeller-induced aerodynamic disturbances, such as downwash and propeller flap, as shown in Fig. 4. Both types exhibit pronounced low-frequency periodic characteristics [15], as indicated by the finite-window analysis of real flight data in Section VI-A.

Therefore, $\boldsymbol{d}(t)$ can be effectively modeled as a combination of an arbitrary finite number of step signals with unknown magnitudes and sinusoidal functions with arbitrary unknown amplitudes and phase angles. Each component $d_i(t), i = 1, \ldots, 6$, takes or can be expanded into the following form

$$d_i(t) = a_{i0} + \sum_{j=1}^{N_i} a_{ij} \sin(\sigma_{ij} t + \eta_{ij}) \tag{7}$$



(a) Downwash disturbance

(b) Propeller flap disturbance

Fig. 4: **Aerodynamic disturbances of propellers.** (a) Downwash: Airflow from the propeller strikes the frame, affecting propeller lift. (b) Flap: Airflow causes unbalanced lift, leading to propeller flapping.

where a_{i0}, a_{ij}, and η_{ij} are unknown magnitudes, amplitudes, and phases, respectively, while σ_{ij} are the identified dominant frequencies. This formulation can effectively capture the dominant deterministic components of a broad class of aerodynamic disturbances. In practice, the dominant frequencies σ_{ij} are estimated from finite-window flight data using the fast Fourier transform (FFT).

Agile quadrotor flight under external disturbances faces two main challenges: high-efficiency time-optimal planning and high-accuracy agile tracking, both of which are affected by model uncertainty and actuator constraints. These challenges can be formulated into the following two key problems.

D. Robust Time-Optimal Trajectory Optimization Problem

Time-optimal waypoint flight aims to minimize the total trajectory time T_h while satisfying the quadrotor dynamics, waypoint constraints, and actuation limits. It is important to note that the solution of this problem must overcome two fundamental challenges: **model uncertainty** and **computational efficiency**. These challenges become more pronounced under external disturbances that lead to conservative solutions.

For an uncertain quadrotor system subject to external disturbances, the robust time-optimal planning problem investigated in this paper can be formulated as follows.

Problem 1. *Given the uncertain quadrotor system* (1)-(4) *with constraints* (6)*, the time-optimal waypoint trajectory planning problem is formulated as follows:*

$$\min_{\boldsymbol{x}_k, \boldsymbol{u}_k, dt} \quad T_h \tag{8a}$$

$$s.t. \quad \|\boldsymbol{p}_{kj} - \boldsymbol{p}_{wj}\|_2^2 \le \delta_j^2, j = 1, \ldots, m \tag{8b}$$

$$\boldsymbol{x}_{\ell+1|k} = f_{RK}(\boldsymbol{x}_{\ell|k}, \boldsymbol{u}_{\ell|k}, dt), \ell \in \mathbb{N}_{0:h-1} \tag{8c}$$

$$\boldsymbol{x}_{\ell|k} \in \mathbb{X}, \ \boldsymbol{u}_{\ell|k} \in \mathbb{U}, \ \textit{and} \ \boldsymbol{x}_0 = \boldsymbol{x}_{init} \tag{8d}$$

where $\boldsymbol{p}_{wj} \in \mathbb{R}^3$ with $j \in [1, \ldots, m]$ denote the positions of m waypoints. $\boldsymbol{p}_{kj}$ and $\delta_j \ge 0$ represent the position of the quadrotor and the permissible position error when passing through the j-th waypoint, respectively. The dynamics constraint is represented in (8c)*, where an explicit 4th-order Runge-Kutta method is used to integrate continuous dynamics*

(1)-(4) *into* h *steps over the trajectory time* T_h*, with a sampling period* $dt = T_h/h$. $\boldsymbol{x}_{\ell|k}$ *and* $\boldsymbol{u}_{\ell|k}$ *denote the prediction of the state and input at time step* $\ell + k$*, made at time* k*, respectively. The constraints on the quadrotor state, the control input, and the initial state are represented in* (8d).

Solving (8) yields the time-optimal reference trajectory:

$$\mathcal{T}^*_{ref} := \{(\boldsymbol{x}^r_k, \dot{\boldsymbol{x}}^r_k, dt^r) \mid k = 1, \dots, h\} \tag{9}$$

where $\boldsymbol{x}^r = \operatorname{col}[\boldsymbol{p}^r, \boldsymbol{v}^r, \boldsymbol{q}^r, \boldsymbol{\omega}^r]$, with $\boldsymbol{x}^r_k \in \mathbb{X}_r := \mathbb{X}^{\boldsymbol{p}^r}_r \times \mathbb{X}^{\boldsymbol{v}^r}_r \times \mathbb{X}^{\boldsymbol{q}^r}_r \times \mathbb{X}^{\boldsymbol{\omega}^r}_r$ and $\dot{\boldsymbol{x}}^r_k \in \mathbb{D}_r := \mathbb{D}^{\boldsymbol{p}^r}_r \times \mathbb{D}^{\boldsymbol{v}^r}_r \times \mathbb{D}^{\boldsymbol{q}^r}_r \times \mathbb{D}^{\boldsymbol{\omega}^r}_r$ for $k = 1, \dots, h$, where $\mathbb{X}_r$ and $\mathbb{D}_r$ are $\mathcal{C}$ sets.

E. Constrained Agile Trajectory Tracking and Disturbance Rejection Problem

Given the time-optimal reference trajectory $\mathcal{T}^*_{ref} := (\boldsymbol{x}^r, \dot{\boldsymbol{x}}^r, t^r)$ where $\boldsymbol{x}^r$ and $\dot{\boldsymbol{x}}^r$ are bounded, the constrained agile trajectory tracking and disturbance rejection problem for the uncertain quadrotor is formulated as follows.

Problem 2. *Given the uncertain quadrotor system* (1)-(4) *with constraints* (6) *and a desired time-optimal trajectory* $\mathcal{T}^*_{ref} : (\boldsymbol{x}^r, \dot{\boldsymbol{x}}^r, t^r)$ *such that* $\boldsymbol{x}^r$ *and* $\dot{\boldsymbol{x}}^r$ *are bounded, design a controller of the form*

$$\begin{aligned} \boldsymbol{u} &= \varrho_1(\boldsymbol{\zeta}_1, \boldsymbol{p}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}, \boldsymbol{p}^r, \boldsymbol{v}^r, \boldsymbol{q}^r, \boldsymbol{\omega}^r, t^r) \\ \dot{\boldsymbol{\zeta}}_1 &= \varphi_1(\boldsymbol{\zeta}_1, \boldsymbol{p}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}, \boldsymbol{p}^r, \boldsymbol{v}^r, \boldsymbol{q}^r, \boldsymbol{\omega}^r) \end{aligned} \tag{10}$$

where $\boldsymbol{\zeta}_1$ *is the compensator state,* ϱ_1 *and* φ_1 *are some sufficiently smooth functions, such that, for any disturbance* $\boldsymbol{d}(t)$ *of the form* (7) *and any initial condition* $\boldsymbol{p}(0), \boldsymbol{v}(0), \boldsymbol{q}(0), \boldsymbol{\omega}(0)$ *and* $\boldsymbol{\zeta}_1(0)$ *under the constraints* (6)*, the solution of the closed-loop system composed of* (1)-(4) *and* (10) *is bounded, satisfies the constraints* (6)*, and*

$$\begin{aligned} &\lim_{t\to\infty}(\boldsymbol{p}(t) - \boldsymbol{p}^r(t)) = \boldsymbol{0}, \quad \lim_{t\to\infty}(\boldsymbol{v}(t) - \boldsymbol{v}^r(t)) = \boldsymbol{0} \\ &\lim_{t\to\infty}(\boldsymbol{q}(t) - \boldsymbol{q}^r(t)) = \boldsymbol{0}, \quad \lim_{t\to\infty}(\boldsymbol{\omega}(t) - \boldsymbol{\omega}^r(t)) = \boldsymbol{0}. \end{aligned} \tag{11}$$

We define the error coordinate transformation as follows:

$$\begin{aligned} \boldsymbol{p}_e &= \boldsymbol{p} - \boldsymbol{p}^r, \qquad \boldsymbol{v}_e = \boldsymbol{v} - \boldsymbol{v}^r \\ \boldsymbol{q}_e &= (\boldsymbol{q}^r)^{-1} \odot \boldsymbol{q}, \quad \boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{B}_e\boldsymbol{\omega}^r. \end{aligned} \tag{12}$$

The unit error quaternion $\boldsymbol{q}_e = \operatorname{col}[q_{ew}, \boldsymbol{q}_{ev}]$ can be put as:

$$\begin{aligned} q_{ew} &= q_w q^r_w + \boldsymbol{q}^\top_v \boldsymbol{q}^r_v \\ \boldsymbol{q}_{ev} &= q^r_w \boldsymbol{q}_v - q_w \boldsymbol{q}^r_v - \boldsymbol{q}^{r\times}_v \boldsymbol{q}_v. \end{aligned} \tag{13}$$

$\boldsymbol{B}_e \in \mathrm{SO}(3)$ is the rotation matrix from the reference attitude frame $\mathcal{R}$ to $\mathcal{B}$, calculated by $\boldsymbol{B}_e = (q^2_{ew} - \boldsymbol{q}^\top_{ev}\boldsymbol{q}_{ev})\boldsymbol{I}_3 + 2\boldsymbol{q}_{ev}\boldsymbol{q}^\top_{ev} - 2q_{ew}\boldsymbol{q}^\times_{ev}$ with $||\boldsymbol{q}_e|| = 1$ and $\dot{\boldsymbol{B}}_e = -\boldsymbol{\omega}^\times_e \boldsymbol{B}_e$.

Define $\boldsymbol{q}_1 = \operatorname{col}[1, 0, 0, 0]$. As in [37], the following coordinate transformation

$$\tilde{\boldsymbol{q}}_e = \boldsymbol{q}_e - \boldsymbol{q}_1 \tag{14}$$

yields

$$\dot{\tilde{\boldsymbol{q}}}_e = \frac{1}{2}\tilde{\boldsymbol{q}}_e \odot q(\boldsymbol{\omega}_e) + \frac{1}{2}\boldsymbol{q}_1 \odot q(\boldsymbol{\omega}_e). \tag{15}$$

Then, the error dynamics can be written as follows:

$$\dot{\boldsymbol{p}}_e = \boldsymbol{v}_e \tag{16a}$$

$$\dot{\boldsymbol{v}}_e = \boldsymbol{g} + f_T\boldsymbol{z}_B + m^{-1}\boldsymbol{d}_f - \dot{\boldsymbol{v}}^r \tag{16b}$$

$$\dot{\tilde{\boldsymbol{q}}}_e = \frac{1}{2}\tilde{\boldsymbol{q}}_e \odot q(\boldsymbol{\omega}_e) + \frac{1}{2}\boldsymbol{q}_1 \odot q(\boldsymbol{\omega}_e) \tag{16c}$$

$$\begin{aligned} J\dot{\boldsymbol{\omega}}_e =& \boldsymbol{\tau} + \boldsymbol{d}_\tau - (\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r)^\times J(\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r) \\ &+ J(\boldsymbol{\omega}^\times_e \boldsymbol{B}_e\boldsymbol{\omega}^r - \boldsymbol{B}_e\dot{\boldsymbol{\omega}}^r). \end{aligned} \tag{16d}$$

Consequently, Problem 2 is transformed into a constrained stabilization problem for the error dynamics (16) under the coordinate transformation (12) and (14). The objective is to design a robust controller of the form (10) such that the original constraints (6) are satisfied and

$$\lim_{t\to\infty}(\boldsymbol{p}_e(t), \boldsymbol{v}_e(t), \tilde{\boldsymbol{q}}_e(t), \boldsymbol{\omega}_e(t)) = (\boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}). \tag{17}$$

Remark 1. *Problem 2 extends standard trajectory tracking problems by involving disturbance rejection, constraint satisfaction, and near-limit operation. Trajectory tracking and disturbance rejection using output regulation have been well studied for several classes of nonlinear systems [14], [17]. However, system constraints are not considered in classic output regulation theory [16], [17], posing considerable challenges for safety and constraint satisfaction during agile flight.*

IV. METHODOLOGY

To address Problem 1 and Problem 2, we present a unified framework for time-optimal trajectory planning and constrained asymptotic tracking control, as illustrated in Fig. 2. First, the internal model compensators are developed in Section IV-A for actively learning and rejecting disturbances. Section IV-B then establishes the theoretical foundation that Problem 2 without constraints can be addressed by an internal model-based controller. Building upon this, Section IV-C constructs a nominal system with tightened constraints, coupled with an adaptive feedback controller to handle parametric uncertainty and restrict deviations between the original and nominal systems. Based on the nominal system, Section IV-D introduces a robust optimal online planner for high-efficiency time-optimal trajectory generation. Finally, Section IV-E presents an optimal controller to regulate the nominal system while satisfying the tightened constraints.

A. Internal Model Design

The quadrotor dynamics (1)-(4) involve two sources of uncertainty: uncertain inertia parameters in J and external disturbances $\boldsymbol{d}(t)$. Since neither the internal model method nor the adaptive control approach can independently stabilize the system under both sources of uncertainty, we first perform a linearization operation on the unknown parameters of the inertia matrix J. As in [38], for any vector $\boldsymbol{\beta} = [\beta_1, \beta_2, \beta_3] \in \mathbb{R}^3$, we have $J\boldsymbol{\beta} = L(\boldsymbol{\beta})[J_{11}\ J_{22}\ J_{33}\ J_{23}\ J_{13}\ J_{12}]^\top$, where

$$L(\boldsymbol{\beta}) = \begin{bmatrix} \beta_1 & 0 & 0 & 0 & \beta_3 & \beta_2 \\ 0 & \beta_2 & 0 & \beta_3 & 0 & \beta_1 \\ 0 & 0 & \beta_3 & \beta_2 & \beta_1 & 0 \end{bmatrix}.$$

If some elements of J are uncertain due to modeling errors or parametric variations, the inertia vector can be linearly parameterized by an unknown vector $\boldsymbol{\gamma}$ as $[J_{11}\ J_{22}\ J_{33}\ J_{23}\ J_{13}\ J_{12}]^\top = \bar{L}_1\boldsymbol{\gamma} + \bar{L}_0$, where $\bar{L}_1 \in \mathbb{R}^{6\times n_\gamma}$ and $\bar{L}_0 \in \mathbb{R}^{6\times 1}$ are some known matrices. Thus,

$$J\boldsymbol{\beta} = L_1(\boldsymbol{\beta})\boldsymbol{\gamma} + L_0(\boldsymbol{\beta}) \tag{18}$$

where $L_1(\boldsymbol{\beta}) = L(\boldsymbol{\beta})\bar{L}_1$ and $L_0(\boldsymbol{\beta}) = L(\boldsymbol{\beta})\bar{L}_0$. Then, we get

$$\begin{aligned} -(\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r)^\times J(\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r) + J(\boldsymbol{\omega}_e^\times \boldsymbol{B}_e\boldsymbol{\omega}^r - \boldsymbol{B}_e\dot{\boldsymbol{\omega}}^r) \\ = \mathcal{F}_1(\boldsymbol{\omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma} + \mathcal{F}_0(\boldsymbol{\omega}_e, \boldsymbol{q}_e) \\ \mathcal{F}_i(\boldsymbol{\omega}_e, \boldsymbol{q}_e) := -(\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r)^\times L_i(\boldsymbol{\omega}_e + \boldsymbol{B}_e\boldsymbol{\omega}^r) + \\ L_i(\boldsymbol{\omega}_e^\times \boldsymbol{B}_e\boldsymbol{\omega}^r - \boldsymbol{B}_e\dot{\boldsymbol{\omega}}^r), i = 0, 1. \end{aligned} \tag{19}$$

Substituting this parameterization into (16d) gives

$$J\dot{\boldsymbol{\omega}}_e = \mathcal{F}_1(\boldsymbol{\omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma} + \mathcal{F}_0(\boldsymbol{\omega}_e, \boldsymbol{q}_e) + \boldsymbol{\tau} + \boldsymbol{d}_\tau. \tag{20}$$

To deal with the external disturbances $d_i(t)$ (for $i = 1, \ldots, 6$), using a standard argument from output regulation theory [16], [17], we obtain that there exist positive integers r_i and real numbers $a_{i,1}, \ldots, a_{i,r_i}$ such that

$$\frac{d^{r_i} d_i}{dt^{r_i}} = a_{i,1}d_i + a_{i,2}\dot{d}_i + \cdots + a_{i,r_i}\frac{d^{r_i-1}d_i}{dt^{r_i-1}}. \tag{21}$$

This is expressed in a compact form by defining

$$\Phi_i = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_{i,1} & a_{i,2} & a_{i,3} & \cdots & a_{i,r_i} \end{bmatrix}, \quad \Psi_i = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^\top \tag{22}$$

and $\boldsymbol{\eta}_i = \mathrm{col}(d_i, \dot{d}_i, \ldots, \frac{d^{r_i-1}d_i}{dt^{r_i-1}})$. Combining (21) and (22), it follows that $\boldsymbol{\eta}_i$ satisfies

$$\frac{d\boldsymbol{\eta}_i}{dt} = \Phi_i\boldsymbol{\eta}_i,\ d_i = \Psi_i\boldsymbol{\eta}_i,\ i = 1, \ldots, 6. \tag{23}$$

Let T_i be any nonsingular matrix of dimension r_i, and

$$\boldsymbol{\theta}_i(t) = -T_i\boldsymbol{\eta}_i(t). \tag{24}$$

Then, by (23) and (24), we get

$$\dot{\boldsymbol{\theta}}_i = T_i\Phi_i T_i^{-1}\boldsymbol{\theta}_i, \quad d_i = -\Psi_i T_i^{-1}\boldsymbol{\theta}_i. \tag{25}$$

Choose a pair of controllable matrices $M_i \in \mathbb{R}^{r_i \times r_i}$ and $N_i \in \mathbb{R}^{r_i}$, where M_i is Hurwitz. Since (M_i, N_i) is controllable, (Φ_i, Ψ_i) is observable, and the spectra of Φ_i and M_i are disjoint, there exists a unique nonsingular matrix T_i that satisfies the Sylvester equation [39]:

$$T_i\Phi_i - M_iT_i = N_i\Psi_i. \tag{26}$$

Following the robust output regulation theory [16], [17], we design two classes of dynamic compensators based on the IMP to reject unknown disturbances in both translation and rotation:

$$\dot{\boldsymbol{\vartheta}}_f = M_f\boldsymbol{\vartheta}_f + N_f\boldsymbol{u}_f - \mathcal{K}_0(\boldsymbol{v}_e) \tag{27a}$$

$$\dot{\boldsymbol{\vartheta}}_\tau = M_\tau\boldsymbol{\vartheta}_\tau + N_\tau\boldsymbol{u}_\tau - \mathcal{K}_1(\boldsymbol{\omega}_e) \tag{27b}$$

where $\boldsymbol{\vartheta}_f \in \mathbb{V}_f \subseteq \mathbb{R}^{r_f}$, $r_f = r_1 + r_2 + r_3$ and $\boldsymbol{\vartheta}_\tau \in \mathbb{V}_\tau \subseteq \mathbb{R}^{r_\tau}$,$r_\tau = r_4 + r_5 + r_6$, $M_f :=$ block diag(M_1, M_2, M_3), $N_f :=$ block diag(N_1, N_2, N_3), $T_f :=$ block diag(T_1, T_2, T_3), $\Phi_f :=$ block diag(Φ_1, Φ_2, Φ_3), $\Psi_f :=$ block diag (Ψ_1, Ψ_2, Ψ_3), $M_\tau :=$ block diag(M_4, M_5, M_6), $N_\tau :=$ block diag(N_4, N_5, N_6), $T_\tau :=$ block diag(T_4, T_5, T_6), $\Phi_\tau :=$ block diag(Φ_4, Φ_5, Φ_6), and $\Psi_\tau :=$ block diag$(\Psi_4,\ \Psi_5, \Psi_6)$. The functions $\mathcal{K}_0(\boldsymbol{v}_e)$ and $\mathcal{K}_1(\boldsymbol{\omega}_e)$ are sufficiently smooth functions to be assigned later.

Finally, we let

$$\boldsymbol{u}_f = f_T\boldsymbol{z}_B + \boldsymbol{g} - \dot{\boldsymbol{v}}^r, \quad \boldsymbol{u}_f \in \bar{\mathbb{U}}_f \subseteq \mathbb{R}^m \tag{28a}$$

$$\boldsymbol{u}_\tau = \boldsymbol{\tau}, \qquad \boldsymbol{u}_\tau \in \bar{\mathbb{U}}_\tau \subseteq \mathbb{R}^m \tag{28b}$$

where

$$\bar{\mathbb{U}}_f = \mathbb{U}_f\boldsymbol{z}_B \oplus \mathbb{G}_f \ominus \mathbb{D}_r^{\boldsymbol{v}^r},\ \bar{\mathbb{U}}_\tau = \mathbb{U}_\tau,\ \mathbb{G}_f = \{\boldsymbol{g}\}. \tag{29}$$

Remark 2. *The dynamic compensators* (27) *are motivated by the IMP [16], [17], which demonstrates remarkable performance in periodic disturbance rejection. Based on the robust output regulation theory, the original agile trajectory tracking and disturbance rejection problem can be converted into an adaptive regulation problem of the augmented system composed of* (16) *and* (27)*.*

We take the following coordinate and input transformation for the translational dynamics:

$$\bar{\boldsymbol{\vartheta}}_f = \boldsymbol{\vartheta}_f - m^{-1}\boldsymbol{\theta}_f, \quad \bar{\boldsymbol{u}}_f = \boldsymbol{u}_f - \Psi_f T_f^{-1}\boldsymbol{\vartheta}_f \tag{30}$$

where $\boldsymbol{\theta}_f := \mathrm{col}(\theta_1, \theta_2, \theta_3)$.

Then, by (16b), (25), (26) and (27a), we have

$$\begin{aligned} \dot{\bar{\boldsymbol{\vartheta}}}_f &= (M_f + N_f\Psi_f T_f^{-1})\bar{\boldsymbol{\vartheta}}_f + N_f\bar{\boldsymbol{u}}_f - \mathcal{K}_0(\boldsymbol{v}_e) \\ \dot{\boldsymbol{v}}_e &= \bar{\boldsymbol{u}}_f + \Psi_f T_f^{-1}\bar{\boldsymbol{\vartheta}}_f. \end{aligned} \tag{31}$$

To regulate the transformed system (31), we define

$$\mathcal{K}_0(\boldsymbol{v}_e) := M_f N_f \boldsymbol{v}_e \tag{32}$$

and take an additional coordinate transformation

$$\tilde{\boldsymbol{\vartheta}}_f = \bar{\boldsymbol{\vartheta}}_f - N_f\boldsymbol{v}_e. \tag{33}$$

Then, by (31), (32) and (33), we obtain

$$\begin{aligned} \dot{\tilde{\boldsymbol{\vartheta}}}_f &= M_f\tilde{\boldsymbol{\vartheta}}_f \\ \dot{\boldsymbol{v}}_e &= \bar{\boldsymbol{u}}_f + \Psi_f T_f^{-1}\tilde{\boldsymbol{\vartheta}}_f + \Psi_f T_f^{-1}N_f\boldsymbol{v}_e. \end{aligned} \tag{34}$$

It is noted that the rotational subsystem in (16) is a nonlinear cascade interconnection [38], which means that the kinematics subsystem (16c) is stabilized only by $\boldsymbol{\omega}_e$ indirectly. Inspired by [38], we further define the following transformation

$$\boldsymbol{\Omega}_e = \boldsymbol{\omega}_e + E_1\boldsymbol{q}_{ev} \tag{35}$$

where $E_1 \in \mathbb{R}^{3\times 3}$ is some symmetric positive definite matrix. Using (20) and (35), we can obtain

$$\begin{aligned} J\dot{\boldsymbol{\Omega}}_e &= \bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma} + \bar{\mathcal{F}}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \boldsymbol{\tau} + \boldsymbol{d}_\tau \\ &\quad + JE_1\left(\frac{1}{2}(q_{ew}\boldsymbol{I}_3 + \boldsymbol{q}_{ev}^\times)\boldsymbol{\omega}_e\right) \\ &= (\bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e))\boldsymbol{\gamma} + \bar{\mathcal{F}}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) \\ &\quad + \mathcal{H}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \boldsymbol{\tau} + \boldsymbol{d}_\tau \end{aligned} \tag{36}$$

where

$$\begin{aligned} \bar{\mathcal{F}}_i(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) &:= \mathcal{F}_i(\boldsymbol{\Omega}_e - E_1\boldsymbol{q}_{ev}, \boldsymbol{q}_e) \\ JE_1\left(\frac{1}{2}(q_{ew}\boldsymbol{I}_3 + \boldsymbol{q}_{ev}^\times)\boldsymbol{\omega}_e\right) &= \mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma} + \mathcal{H}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) \\ \mathcal{H}_i(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) &:= L_i\left(\frac{1}{2}E_1(q_{ew}\boldsymbol{I}_3 + \boldsymbol{q}_{ev}^\times)\boldsymbol{\omega}_e\right), i = 0, 1. \end{aligned} \tag{37}$$

To achieve adaptive regulation, the following coordinate and input transformation is introduced for the rotational dynamics

$$\bar{\boldsymbol{\vartheta}}_\tau = \boldsymbol{\vartheta}_\tau - \boldsymbol{\theta}_\tau, \quad \bar{\boldsymbol{u}}_\tau = \boldsymbol{u}_\tau - \Psi_\tau T_\tau^{-1}\boldsymbol{\vartheta}_\tau \tag{38}$$

where $\boldsymbol{\theta}_\tau := \text{col}(\theta_4, \theta_5, \theta_6)$. Then, by (25), (26), (27b) and (36), we obtain

$$\begin{aligned}\dot{\bar{\boldsymbol{\vartheta}}}_\tau =&(M_\tau + N_\tau\Psi_\tau T_\tau^{-1})\bar{\boldsymbol{\vartheta}}_\tau + N_\tau\bar{\boldsymbol{u}}_\tau - \mathcal{K}_1(\boldsymbol{\omega}_e)\\ J\dot{\boldsymbol{\Omega}}_e =&(\bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e))\boldsymbol{\gamma} + \bar{\boldsymbol{u}}_\tau + \Psi_\tau T_\tau^{-1}\bar{\boldsymbol{\vartheta}}_\tau\\ &+ \bar{\mathcal{F}}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e).\end{aligned} \tag{39}$$

To regulate the transformed system (39), we let

$$\mathcal{K}_1(\boldsymbol{\omega}_e) := M_\tau N_\tau \bar{L}_0(\boldsymbol{\Omega}_e) \tag{40}$$

where

$$\bar{L}_i(\boldsymbol{\Omega}_e) := L_i(\boldsymbol{\Omega}_e - E_1\boldsymbol{q}_{ev}), i = 0, 1. \tag{41}$$

Then, an additional transformation is defined as follows

$$\check{\boldsymbol{\vartheta}}_\tau = \bar{\boldsymbol{\vartheta}}_\tau - N_\tau J\boldsymbol{\Omega}_e. \tag{42}$$

By (39), (40) and (42), the following dynamics are obtained

$$\begin{aligned}\dot{\check{\boldsymbol{\vartheta}}}_\tau =&M_\tau\check{\boldsymbol{\vartheta}}_\tau + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma}\\ J\dot{\boldsymbol{\Omega}}_e =&\left(\bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{C}_1(\boldsymbol{\Omega}_e)\right)\boldsymbol{\gamma} + \bar{\boldsymbol{u}}_\tau\\ &+ \Psi_\tau T_\tau^{-1}\check{\boldsymbol{\vartheta}}_\tau + \bar{\mathcal{F}}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{C}_0(\boldsymbol{\Omega}_e)\end{aligned} \tag{43}$$

where

$$\begin{aligned}\mathcal{C}_i(\boldsymbol{\Omega}_e) =& \Psi_\tau T_\tau^{-1} N_\tau \bar{L}_i(\boldsymbol{\Omega}_e), i = 0, 1\\ \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) =& M_\tau N_\tau \bar{L}_1(\boldsymbol{\Omega}_e) - N_\tau\bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) - N_\tau\mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e).\end{aligned}$$

Furthermore, we design an auxiliary system to handle the parametric uncertainty $\boldsymbol{\gamma}$ in (43) as follows:

$$\dot{\boldsymbol{z}} = M_\tau\boldsymbol{z} + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) \tag{44}$$

where $\boldsymbol{z} \in \mathbb{R}^{r_\tau \times n_\gamma}$. We define the coordinate transformation as follows:

$$\tilde{\boldsymbol{\vartheta}}_\tau = \check{\boldsymbol{\vartheta}}_\tau - \boldsymbol{z}\boldsymbol{\gamma}. \tag{45}$$

Then, by (43) and (44), we get

$$\begin{aligned}\dot{\tilde{\boldsymbol{\vartheta}}}_\tau =& M_\tau\check{\boldsymbol{\vartheta}}_\tau + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\boldsymbol{\gamma} - (M_\tau\boldsymbol{z} + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e))\boldsymbol{\gamma}\\ =& M_\tau\tilde{\boldsymbol{\vartheta}}_\tau\\ J\dot{\boldsymbol{\Omega}}_e =& \rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\boldsymbol{\gamma} + \bar{\boldsymbol{u}}_\tau + \Psi_\tau T_\tau^{-1}\tilde{\boldsymbol{\vartheta}}_\tau + g(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\end{aligned} \tag{46}$$

where

$$\begin{aligned}\rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z}) =& \bar{\mathcal{F}}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_1(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{C}_1(\boldsymbol{\Omega}_e) + \Psi_\tau T_\tau^{-1}\boldsymbol{z}\\ g(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) =& \bar{\mathcal{F}}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_0(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{C}_0(\boldsymbol{\Omega}_e).\end{aligned} \tag{47}$$

Then, the augmented system can be obtained:

$$\dot{\boldsymbol{p}}_e = \boldsymbol{v}_e \tag{48a}$$

$$\dot{\boldsymbol{v}}_e = \bar{\boldsymbol{u}}_f + \Psi_f T_f^{-1}\tilde{\boldsymbol{\vartheta}}_f + \Psi_f T_f^{-1} N_f\boldsymbol{v}_e \tag{48b}$$

$$\dot{\tilde{\boldsymbol{q}}}_e = \frac{1}{2}\tilde{\boldsymbol{q}}_e \odot q(\boldsymbol{\Omega}_e - E_1\boldsymbol{q}_{ev}) + \frac{1}{2}\boldsymbol{q}_1 \odot q(\boldsymbol{\Omega}_e - E_1\boldsymbol{q}_{ev}) \tag{48c}$$

$$J\dot{\boldsymbol{\Omega}}_e = \rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\boldsymbol{\gamma} + \bar{\boldsymbol{u}}_\tau + \Psi_\tau T_\tau^{-1}\tilde{\boldsymbol{\vartheta}}_\tau + g(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) \tag{48d}$$

$$\dot{\tilde{\boldsymbol{\vartheta}}}_f = M_f\tilde{\boldsymbol{\vartheta}}_f \tag{48e}$$

$$\dot{\tilde{\boldsymbol{\vartheta}}}_\tau = M_\tau\tilde{\boldsymbol{\vartheta}}_\tau \tag{48f}$$

$$\dot{\boldsymbol{z}} = M_\tau\boldsymbol{z} + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e). \tag{48g}$$

The augmented system state $\tilde{\boldsymbol{x}}$ is denoted by the following:

$$\tilde{\boldsymbol{x}} = \text{col}\left[\boldsymbol{p}_e, \boldsymbol{v}_e, \tilde{\boldsymbol{q}}_e, \boldsymbol{\Omega}_e, \tilde{\boldsymbol{\vartheta}}_f, \tilde{\boldsymbol{\vartheta}}_\tau, \boldsymbol{z}\right]. \tag{49}$$

Remark 3. *By introducing the auxiliary system* (44) *and transformation* (30)*,* (33)*,* (35)*,* (38)*,* (42) *and* (45)*, the adaptive regulation problem of the system composed of* (16) *and* (27) *can be converted into an adaptive stabilization problem for system* (48)*. However, it is important to note that the system constraints* (6) *and constraints reconstruction in* (30) *and* (38) *are not considered in classical output regulation theory.*

We denote the transformed control input as $\bar{\boldsymbol{u}} = \text{col}\,[\bar{\boldsymbol{u}}_f, \bar{\boldsymbol{u}}_\tau]$. For $\bar{\boldsymbol{u}}_f \in \mathbb{Z}_f$ and $\bar{\boldsymbol{u}}_\tau \in \mathbb{Z}_\tau$, there exists a compact and convex set $\mathbb{Z}$ containing the origin such that $\bar{\boldsymbol{u}} \in \mathbb{Z}$.

Now, the agile trajectory tracking and disturbance rejection problem of the uncertain quadrotor without constraints (6) can be converted to an adaptive stabilization problem of the augmented system (48). This result is summarized as follows.

Lemma 1. *Consider the augmented system* (48)*. Suppose that there exists a controller of the form*

$$\begin{aligned}\bar{\boldsymbol{u}} =& \varrho_2(\varsigma, \boldsymbol{z}, \boldsymbol{p}_e, \boldsymbol{v}_e, \tilde{\boldsymbol{q}}_e, \boldsymbol{\Omega}_e, t^r)\\ \dot{\varsigma} =& \varphi_2(\varsigma, \boldsymbol{z}, \tilde{\boldsymbol{q}}_e, \boldsymbol{\Omega}_e)\end{aligned} \tag{50}$$

where ϱ_2 and φ_2 are some sufficiently smooth functions, such that the solution of the resulting closed-loop system (48) *is bounded and* $\lim_{t\to\infty}(\boldsymbol{p}_e(t), \boldsymbol{v}_e(t), \tilde{\boldsymbol{q}}_e(t), \boldsymbol{\Omega}_e(t)) = (\boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0})$*. Then, the agile trajectory tracking and disturbance rejection problem without constraints* (6) *is solved.*

The proof of Lemma 1 is provided in Appendix VIII-A.

Remark 4. *By Lemma 1, the agile trajectory tracking and disturbance rejection problem without constraints is converted into an adaptive stabilization problem of the augmented system* (48)*. Accordingly, we design a feedback controller to stabilize* (48) *without considering the original system constraints* (6)*.*

B. Agile Trajectory Tracking and Disturbance Rejection without Constraints

Before formulating the control scheme, we first investigate the stability property obtained through a virtual control design, which provides an auxiliary solution for stabilizing the augmented system. The virtual controller is designed as follows:

$$\begin{aligned}\bar{\boldsymbol{u}} =& \begin{bmatrix}\bar{\boldsymbol{u}}_f\\ \bar{\boldsymbol{u}}_\tau\end{bmatrix} = \begin{bmatrix}-k_p\boldsymbol{p}_e - K_v\boldsymbol{v}_e - \Psi_f T_f^{-1} N_f\boldsymbol{v}_e\\ -k_q\boldsymbol{q}_{ev} - K_\Omega\boldsymbol{\Omega}_e - \rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\hat{\boldsymbol{\gamma}} - g(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)\end{bmatrix}\\ \dot{\hat{\boldsymbol{\gamma}}} =& \Lambda_1\rho^\top(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\boldsymbol{\Omega}_e\end{aligned} \tag{51}$$

where k_p and k_q are some positive constants, $K_v \in \mathbb{R}^{3\times3}$ and $K_\Omega \in \mathbb{R}^{3\times3}$ are some symmetric positive definite matrices, and $\lambda_m(K_v) > 1$, $\lambda_m(K_\Omega) > 1$. $\hat{\boldsymbol{\gamma}}$ is an estimate of $\boldsymbol{\gamma}$ and $\Lambda_1 \in \mathbb{R}^{n_\gamma \times n_\gamma}$ is an arbitrary positive definite constant matrix.

The next result characterizes the closed-loop stability.

Theorem 1. *The agile trajectory tracking and disturbance rejection Problem 2, without considering the constraints* (6)*, can be solved by the following internal model controller:*

$$
\begin{aligned}
\boldsymbol{u} &= \begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} (\bar{\boldsymbol{u}}_f + \Psi_f T_f^{-1}\boldsymbol{\vartheta}_f)\cdot \boldsymbol{z}_B \\ \bar{\boldsymbol{u}}_\tau + \Psi_\tau T_\tau^{-1}\boldsymbol{\vartheta}_\tau \end{bmatrix} \\
\dot{\boldsymbol{\vartheta}} &= \begin{bmatrix} \dot{\boldsymbol{\vartheta}}_f \\ \dot{\boldsymbol{\vartheta}}_\tau \end{bmatrix} = \begin{bmatrix} M_f\boldsymbol{\vartheta}_f + N_f\boldsymbol{u}_f - M_f N_f \boldsymbol{v}_e \\ M_\tau\boldsymbol{\vartheta}_\tau + N_\tau\boldsymbol{u}_\tau - M_\tau N_\tau \bar{L}_0(\boldsymbol{\Omega}_e) \end{bmatrix} \\
\dot{\tilde{\boldsymbol{\gamma}}} &= \Lambda_1 \rho^\top(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\boldsymbol{\Omega}_e \\
\dot{\boldsymbol{z}} &= M_\tau \boldsymbol{z} + \xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)
\end{aligned} \tag{52}
$$

where $\bar{\boldsymbol{u}}_i$ is a virtual controller defined in (51)*, $\Psi_i T_i^{-1}$ is defined in* (25)*, M_i and N_i are defined in* (26) *with $i \in \{f,\tau\}$, $g(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)$ is defined in* (47)*, $\bar{L}_0(\boldsymbol{\Omega}_e)$ is defined in* (41)*, Λ_1 is defined in* (51)*, $\rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})$ is defined in* (47)*, and $\xi(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)$ is defined in* (43)*.*

The proof of Theorem 1 is provided in Appendix VIII-B.

Remark 5. *By Theorem 1, there exists a controller composed of the internal models* (27) *and an adaptive feedback stabilizer* (51) *that solves Problem 2 without constraints* (6)*. This step serves as the theoretical basis, proving that the core challenge caused by parametric uncertainty and external disturbances can be addressed in principle. In the following, we develop a comprehensive control strategy to address the constrained agile trajectory tracking and disturbance rejection Problem 2.*

C. Nominal Dynamics Design and Constraints Reconstruction

This section presents the core mechanism enabling robust optimal planning and control: the systematic conversion of the original uncertain quadrotor system (1)-(4) with constraints (6) into a nominal system with tightened constraints. The conversion proceeds using a two-stage transformation: 1) dynamics decoupling via an internal model-based adaptive feedback controller, and 2) constraint reconstruction that ensures robust satisfaction of the original constraints.

Since the original system (1)-(4) contains parametric uncertainty and unknown disturbances, directly using it in optimal planning and control leads to an uncertain optimization problem. Thus, we construct a disturbance-decoupled nominal system for both time-optimal planning and NMPC stabilization:

$$
\begin{aligned}
&\dot{\hat{\boldsymbol{p}}} = \hat{\boldsymbol{v}}, \qquad \dot{\hat{\boldsymbol{v}}} = \boldsymbol{g} + \hat{\boldsymbol{u}}_f \\
&\dot{\hat{\boldsymbol{q}}} = \frac{1}{2}\hat{\boldsymbol{q}} \odot q(\hat{\boldsymbol{\omega}}), \quad J\dot{\hat{\boldsymbol{\omega}}} = \mathcal{G}_1(\hat{\boldsymbol{\omega}})\boldsymbol{\gamma} + \mathcal{G}_0(\hat{\boldsymbol{\omega}}) + \hat{\boldsymbol{u}}_\tau
\end{aligned} \tag{53}
$$

where

$$
\begin{aligned}
-\hat{\boldsymbol{\omega}}^\times J\hat{\boldsymbol{\omega}} &= \mathcal{G}_1(\hat{\boldsymbol{\omega}})\boldsymbol{\gamma} + \mathcal{G}_0(\hat{\boldsymbol{\omega}}) \\
\mathcal{G}_i(\hat{\boldsymbol{\omega}}) &:= -\hat{\boldsymbol{\omega}}^\times L_i(\hat{\boldsymbol{\omega}}), i = 0, 1.
\end{aligned} \tag{54}
$$

The nominal system state $\hat{\boldsymbol{x}}$ and the control input $\hat{\boldsymbol{u}}$ are denoted by the following:

$$
\hat{\boldsymbol{x}} = \mathrm{col}\left[\hat{\boldsymbol{p}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{q}}, \hat{\boldsymbol{\omega}}\right], \quad \hat{\boldsymbol{u}} = \mathrm{col}\left[\hat{\boldsymbol{u}}_f, \hat{\boldsymbol{u}}_\tau\right]. \tag{55}
$$

Then, the system errors are defined as follows:

$$
\begin{aligned}
\boldsymbol{e}_1 &= \boldsymbol{p} - \hat{\boldsymbol{p}}, \qquad \boldsymbol{e}_2 = \boldsymbol{v} - \hat{\boldsymbol{v}} \\
\boldsymbol{e}_3 &= \hat{\boldsymbol{q}}^{-1} \odot \boldsymbol{q}, \quad \boldsymbol{e}_4 = \boldsymbol{\omega} - \hat{\boldsymbol{\omega}}.
\end{aligned} \tag{56}
$$

As in (14) and (35), we introduce the same transformation:

$$
\tilde{\boldsymbol{e}}_3 = \boldsymbol{e}_3 - \boldsymbol{q}_1, \quad \tilde{\boldsymbol{e}}_4 = \boldsymbol{e}_4 + E_2\boldsymbol{e}_{3v} \tag{57}
$$

where $E_2 \in \mathbb{R}^{3\times 3}$ is some symmetric positive definite matrix, $\tilde{\boldsymbol{e}}_3 = \mathrm{col}[\tilde{e}_{3w}, \tilde{\boldsymbol{e}}_{3v}]$, and $\tilde{\boldsymbol{e}}_{3v} = \boldsymbol{e}_{3v}$. By the original system (1)-(4), its corresponding nominal system (53), and the system errors (56), (57), the error dynamics can be written as follows:

$$
\begin{aligned}
\dot{\boldsymbol{e}}_1 =& \boldsymbol{e}_2 \\
\dot{\boldsymbol{e}}_2 =& f_T\boldsymbol{z}_B + m^{-1}\boldsymbol{d}_f - \hat{\boldsymbol{u}}_f \\
\dot{\tilde{\boldsymbol{e}}}_3 =& \frac{1}{2}\tilde{\boldsymbol{e}}_3 \odot q(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v}) + \frac{1}{2}\boldsymbol{q}_1 \odot q(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v}) \\
J\dot{\tilde{\boldsymbol{e}}}_4 =& \boldsymbol{\tau} + \boldsymbol{d}_\tau - \hat{\boldsymbol{u}}_\tau + (\mathcal{G}_1(\boldsymbol{\omega}) - \mathcal{G}_1(\hat{\boldsymbol{\omega}}) + \mathcal{H}_1(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4))\boldsymbol{\gamma} \\
&+ \mathcal{G}_0(\boldsymbol{\omega}) - \mathcal{G}_0(\hat{\boldsymbol{\omega}}) + \mathcal{H}_0(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) \\
=& \boldsymbol{\tau} + \boldsymbol{d}_\tau - \hat{\boldsymbol{u}}_\tau + \kappa_1(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4)\boldsymbol{\gamma} + \kappa_0(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4)
\end{aligned} \tag{58}
$$

where, for $i \in \{0, 1\}$,

$$
\begin{aligned}
&\kappa_i(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) = \mathcal{G}_i(\boldsymbol{\omega}) - \mathcal{G}_i(\hat{\boldsymbol{\omega}}) + \mathcal{H}_i(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) \\
&= -(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v})^\times L_i(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v}) - (\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v})^\times L_i(\hat{\boldsymbol{\omega}}) \\
&\quad -\hat{\boldsymbol{\omega}}^\times L_i(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v}) + \mathcal{H}_i(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) \\
&\mathcal{H}_i(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) := L_i\left(\frac{1}{2}E_2((1+\tilde{e}_{3w})\boldsymbol{I}_3 + \tilde{\boldsymbol{e}}_{3v}^\times)(\tilde{\boldsymbol{e}}_4 - E_2\tilde{\boldsymbol{e}}_{3v})\right).
\end{aligned} \tag{59}
$$

We denote $\boldsymbol{x}_e = \mathrm{col}\left[\boldsymbol{e}_1, \boldsymbol{e}_2, \tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4\right]$ as the state of the error dynamics. To guarantee that this nominal system (53) can replace the original system, we design the adaptive controller for the original system (1)-(4) as

$$
\begin{aligned}
\boldsymbol{u} &= \begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} (\hat{\boldsymbol{u}}_f + \Psi_f T_f^{-1}\boldsymbol{\vartheta}_f + \boldsymbol{\delta}_f(\boldsymbol{e}_1, \boldsymbol{e}_2))\boldsymbol{z}_B \\ \hat{\boldsymbol{u}}_\tau + \Psi_\tau T_\tau^{-1}\boldsymbol{\vartheta}_\tau + \boldsymbol{\delta}_\tau(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z}) \end{bmatrix} \\
\dot{\tilde{\boldsymbol{\gamma}}} &= \Lambda_2 \bar{\kappa}_1^\top(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z})\tilde{\boldsymbol{e}}_4
\end{aligned} \tag{60}
$$

where

$$
\begin{aligned}
\boldsymbol{\delta}_f(\boldsymbol{e}_1, \boldsymbol{e}_2) &= -k_1\boldsymbol{e}_1 - K_2\boldsymbol{e}_2 \\
\bar{\kappa}_1(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z}) &= \kappa_1(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4) + \Psi_\tau T_\tau^{-1}\boldsymbol{z} \\
\boldsymbol{\delta}_\tau(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z}) &= -k_3\tilde{\boldsymbol{e}}_{3v} - K_4\tilde{\boldsymbol{e}}_4 - \bar{\kappa}_1(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z})\hat{\boldsymbol{\gamma}} - \kappa_0(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4)
\end{aligned}
$$

k_1 and k_3 are some positive constants, $K_2 \in \mathbb{R}^{3\times 3}$ and $K_4 \in \mathbb{R}^{3\times 3}$ are some symmetric positive definite matrices. It can be verified $\boldsymbol{\delta}_f(\boldsymbol{e}_1, \boldsymbol{e}_2)$ and $\boldsymbol{\delta}_\tau(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z})$ are smooth functions vanishing at their origins, respectively.

We first establish one technical lemma as follows.

Lemma 2. *Consider a positive-definite function $W(t) = \boldsymbol{\mu}^\top(t)\boldsymbol{\mu}(t) + \boldsymbol{\nu}^\top(t)\boldsymbol{\nu}(t)$, where $\dot{\boldsymbol{\mu}}(t) = \psi_1(\boldsymbol{\nu}(t), \boldsymbol{\varpi}(t))$, and $\dot{\boldsymbol{\nu}}(t) = \psi_2(\boldsymbol{\nu}(t), \boldsymbol{\varpi}(t))$. Moreover, $\boldsymbol{\varpi}(t)$ is continuously differentiable, bounded and satisfies $\lim_{t\to\infty}\boldsymbol{\varpi}(t) = \mathbf{0}$. If the time derivative of $W(t)$ satisfies the inequality*

$$
\dot{W}(t) \le -k\|\boldsymbol{\mu}(t)\|^2 + a\|\boldsymbol{\varpi}(t)\|^2 \tag{61}
$$

where k and a are positive constants, then $W(t)$ is bounded.

The proof of Lemma 2 is provided in Appendix VIII-C.

Next, by Lemma 2, we show the feasibility of the replacement of system (1)-(4) by the nominal system (53).

Theorem 2. *With the adaptive feedback controller* (60)*, the state $\hat{\boldsymbol{x}}$ of the nominal system* (53) *asymptotically converges to the state $\boldsymbol{x}$ of the original system* (1)-(4) *as $t \to \infty$, that is,*

$$
\lim_{t\to\infty}(\boldsymbol{e}_1(t), \boldsymbol{e}_2(t), \tilde{\boldsymbol{e}}_3(t), \tilde{\boldsymbol{e}}_4(t)) = (\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}). \tag{62}
$$

The proof of Theorem 2 is provided in Appendix VIII-D.

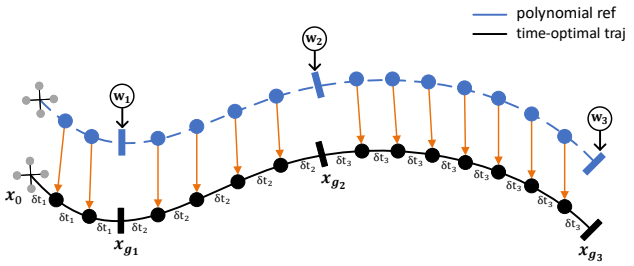


Fig. 5: **Polynomial-based waypoint allocation**. Top: A time-parameterized polynomial trajectory through the next ϕ waypoints is generated from the current state; discrete points of each segment are then mapped to the corresponding time-optimal segments. Bottom: Each segment is assigned a separate sample interval δt_j, and each waypoint constraint is imposed at cumulative discretization index g_j.

By Theorem 2, we have $\lim_{t\to\infty} \boldsymbol{\delta}_f(\boldsymbol{e}_1, \boldsymbol{e}_2) = \mathbf{0}$ and $\lim_{t\to\infty} \boldsymbol{\delta}_\tau(\tilde{\boldsymbol{e}}_3, \tilde{\boldsymbol{e}}_4, \boldsymbol{z}) = \mathbf{0}$. Thus, there exist some compact sets Δ_f and Δ_τ containing the origin such that

$$\boldsymbol{\delta}_f(t) \in \Delta_f,\ \boldsymbol{\delta}_\tau(t) \in \Delta_\tau,\ \forall t > 0. \tag{63}$$

To ensure that the original system (1)-(4) satisfies (6), the constraints of the nominal system (53) are now reconstructed. By Theorem 2, $\boldsymbol{x}_e(t)$ is bounded and converges to the origin. It follows from the proof of Theorem 2 that there always exists a positive invariant set $\mathbb{V}_0$ such that $\boldsymbol{x}_e(t) \in \mathbb{V}_0$ for all $t \geq 0$.

By (30), (38) and (63), the contractive constraints of the nominal system (53) are designed by

$$\hat{\boldsymbol{x}} \in \hat{\mathbb{X}} := \mathbb{X} \ominus \mathbb{V}_0,\ \hat{\boldsymbol{u}} \in \hat{\mathbb{Z}}_i := \mathbb{Z}_i \ominus \Delta_i, i \in \{f, \tau\} \tag{64}$$

where $\mathbb{Z}_i := \bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1} \mathbb{V}_i$.

By Theorem 2, we show that the nominal system (53) with tightened constraints (64) is equivalent to the original system (1)-(4) with constraints (6). With this equivalence, we systematically decouple the uncertain dynamics from the optimization loop. Now, we proceed to solve the robust time-optimal trajectory optimization Problem 1 based on the nominal system (53) with tightened constraints (64).

D. Robust Time-Optimal Online Trajectory Planner

To adapt to multiple uncertainties and large-scale, unforeseeable environments on-the-fly, the time-optimal trajectory optimization problem faces two main challenges.

1) Real-time feasibility: Problem 1 requires reconciling the total trajectory time with the waypoint time allocation in a specified sequence. Such allocation is unknown a priori, rendering the optimization problem challenging to address.

2) Model uncertainty: State-of-the-art time-optimal planners usually rely on simplified linear first-order rotor-drag models to approximate aerodynamic effects [6], which are inaccurate during agile flight or under external disturbances.

To address the first challenge, we develop a receding-horizon polynomial-based waypoint allocation method. At each planning step, the next ϕ waypoints are considered, and a ϕ-segment polynomial trajectory is generated from the current quadrotor state to provide waypoint-index assignment and segment-wise timing initialization. Specifically, each segment is represented by a third-order polynomial:

$$\boldsymbol{\mu}_i(t) = \boldsymbol{a}_i^\top \boldsymbol{\beta}(t), \quad \forall t \in [0, \bar{T}_i] \tag{65}$$

where $\boldsymbol{a}_i^\top = [a_0, a_1, a_2, a_3]$ and $\boldsymbol{\beta}(t) = \text{col}[1, t, t^2, t^3]$. This trajectory ensures position and velocity continuity at each waypoint, with the velocity direction aligned with the corresponding gate normal. The initial segment duration $\bar{T}_i$ is estimated via a bang-bang point-mass model:

$$\bar{T}_i = \max\left(2\sqrt{\frac{\kappa_i \|\boldsymbol{p}_{w,i+1} - \boldsymbol{p}_{w,i}\|}{a_{\max}}}, \frac{\kappa_i \|\boldsymbol{p}_{w,i+1} - \boldsymbol{p}_{w,i}\|}{v_{\text{nom}}}\right) \tag{66}$$

where $a_{\max}$ is the maximum acceleration and v_{nom} is a nominal cruising speed. The factor $\kappa_i = 1 + \alpha(1 - \boldsymbol{n}_i \cdot \boldsymbol{n}_{i+1}),\ \alpha > 0$, is an orientation-aware heuristic correction, where $\boldsymbol{n}_i$ denotes the unit normal vector of the i-th gate. We set $\alpha = 0.5$ for all experiments without track-specific tuning.

The polynomial trajectory (65) is then used to pre-assign each waypoint to a specific discrete state. Let $g_j \in \mathbb{N}$, with $g_0 = 0$, denote the cumulative number of discretization points up to the j-th waypoint. The waypoint constraint is then assigned to the state $\boldsymbol{x}_{g_j}$ (a specific state at the g_j-th discretization point). Thus, the original mixed-integer time-allocation problem is converted into a tractable nonlinear optimization, where each segment is allocated a separate time interval δt_j. The total flight time over the receding-horizon window is:

$$T_\phi = \sum_{j=1}^{\phi} (g_j - g_{j-1}) \delta t_j. \tag{67}$$

As illustrated in Fig. 5, the polynomial provides a spatially close initial approximation and an efficient warm start for the subsequent optimization. Importantly, this polynomial module only determines the initial waypoint-index assignment and segment-wise timing initialization. The final reference is obtained by solving the nonlinear time-optimal problem. Thus, possible conservatism of the heuristic, such as in distant but oppositely oriented gates, is confined to initialization.

To address the second challenge, the time-optimal optimization is formulated on the nominal dynamics (53) with tightened constraints (64). The resulting receding-horizon time-optimal trajectory generation problem is:

$$\min_{\hat{\boldsymbol{x}}_{g_i}, \hat{\boldsymbol{u}}_{g_i}, \delta t_j} \quad T_\phi \tag{68a}$$

$$\text{s.t.} \quad \|\hat{\boldsymbol{p}}_{g_j} - \boldsymbol{p}_{wj}\|_2^2 \leq \varepsilon_j^2, j = 1, \ldots, \phi \tag{68b}$$

$$\hat{\boldsymbol{x}}_{g_i+1} = f_{RK}(\hat{\boldsymbol{x}}_{g_i}, \hat{\boldsymbol{u}}_{g_i}, \delta t_j) \tag{68c}$$

$$\hat{\boldsymbol{x}}_{g_i} \in \hat{\mathbb{X}},\ \hat{\boldsymbol{u}}_{g_i} \in \hat{\mathbb{Z}},\ \text{and}\ \hat{\boldsymbol{x}}_0 = \boldsymbol{x}_{init} \tag{68d}$$

where $g_i \in \mathbb{N}$, with $g_{j-1} \leq g_i \leq g_j$, denotes the discrete points within the j-th trajectory segment, and ϕ is the number of upcoming waypoints considered in the receding-horizon window. The final time-optimal trajectory is obtained as:

$$\mathcal{T}_{ref}^* := \{(\hat{\boldsymbol{x}}_{g_i}^r, \dot{\hat{\boldsymbol{x}}}_{g_i}^r, \delta t_j^r) | g_{j-1} \leq g_i \leq g_j, j = 1, \ldots, \phi\} \tag{69}$$

where $\hat{\boldsymbol{x}}^r = \text{col}\,[\hat{\boldsymbol{p}}^r, \hat{\boldsymbol{v}}^r, \hat{\boldsymbol{q}}^r, \hat{\boldsymbol{\omega}}^r]$.

E. Agile Trajectory Tracking and Disturbance Rejection with Constraints

In this section, a comprehensive framework is established to address the constrained agile trajectory tracking and disturbance rejection Problem 2 for an uncertain quadrotor system, which consists of two classes of nonlinear internal models and a nominal system-based NMPC optimization stabilizer.

To start with, we define an optimization problem within the NMPC framework based on Theorems 1-2. This problem aims to stabilize the nominal system (53) with reconstructed tightened constraints (64), which ensures the original quadrotor system (1)-(4) satisfies the constraints (6).

We define the cost vector as

$$\boldsymbol{y} = \mathrm{col}\left(\hat{\boldsymbol{p}} - \hat{\boldsymbol{p}}^r, \hat{\boldsymbol{v}} - \hat{\boldsymbol{v}}^r, \hat{\boldsymbol{q}} - \hat{\boldsymbol{q}}^r, \hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}}^r\right). \tag{70}$$

Then, at time step k, the NMPC law is set up as solving the following finite-time optimization problem for the nominal system (53) with a receding horizon N:

$$\min_{\hat{\boldsymbol{u}}_{\ell|k}, \hat{\boldsymbol{x}}_k} J(\boldsymbol{y}_k, \hat{\boldsymbol{u}}_k) = l_T(\boldsymbol{y}_{N|k}) + \sum_{\ell=0}^{N-1} l(\boldsymbol{y}_{\ell|k}, \hat{\boldsymbol{u}}_{\ell|k}) \tag{71a}$$

$$\text{s.t.} \quad \hat{\boldsymbol{x}}_{\ell+1|k} = f_{RK}(\hat{\boldsymbol{x}}_{\ell|k}, \hat{\boldsymbol{u}}_{\ell|k}, \delta t_j^r) \tag{71b}$$

$$\hat{\boldsymbol{x}}_{\ell|k} \in \hat{\mathbb{X}}, \quad \hat{\boldsymbol{u}}_{\ell|k} \in \hat{\mathbb{Z}}, \ell \in \mathbb{N}_{0:N-1} \tag{71c}$$

$$\hat{\boldsymbol{x}}_{0|k} = \boldsymbol{x}_{init}, \ \boldsymbol{y}_{N|k} \in \mathbb{T}_f. \tag{71d}$$

The objective function (71a) consists of two components: a running objective $l(\boldsymbol{y}_k, \hat{\boldsymbol{u}}_k) = \|\boldsymbol{y}_k\|_Q^2 + \|\hat{\boldsymbol{u}}_k\|_R^2$ and a terminal cost $l_T(\boldsymbol{y}_k) = \|\boldsymbol{y}_k\|_{Q_T}^2$. The matrices Q, Q_T and R are all positive definite. The terminal constraint $\mathbb{T}_f$ is a closed set containing the origin. The dynamic constraints are represented in (71b). We substitute the nominal dynamics (53) in the Runge-Kutta integration, with the optimal sampling period δt_j^r corresponding to each segment.

By solving (71), we obtain the optimal solutions:

$$\begin{aligned} \hat{\boldsymbol{u}}_k^* &= \{\hat{\boldsymbol{u}}_{0|k}^*, \hat{\boldsymbol{u}}_{1|k}^*, \ldots, \hat{\boldsymbol{u}}_{N-1|k}^*\} \\ \hat{\boldsymbol{x}}_k^* &= \{\hat{\boldsymbol{x}}_{0|k}^*, \hat{\boldsymbol{x}}_{1|k}^*, \ldots, \hat{\boldsymbol{x}}_{N|k}^*\}. \end{aligned} \tag{72}$$

The first action of $\hat{\boldsymbol{u}}_k^*$ is then applied to the nominal system (53), yielding the NMPC policy at time step k:

$$\boldsymbol{\mu}_{nmpc}(\hat{\boldsymbol{x}}_k) = \hat{\boldsymbol{u}}_{0|k}^* = \begin{bmatrix} \hat{\boldsymbol{u}}_{f,0|k}^* \\ \hat{\boldsymbol{u}}_{\tau,0|k}^* \end{bmatrix}. \tag{73}$$

Then, we establish the recursive feasibility of the optimization (71) and the stability of the nominal system (53), thereby providing the solution to Problem 2. Before proceeding, the following definition and standard assumption are necessary.

Definition 1. *A set $\mathbb{X}$ is referred to as a positive invariant set for the system $x_{k+1} = f(x_k)$ if, for all $x_k \in \mathbb{X}$, $k \in \mathbb{N}$, it holds that $x_{k+1} \in \mathbb{X}$.*

Assumption 1. *There exists an appropriate controller $\kappa_f(\boldsymbol{y}_k) \in \hat{\mathbb{Z}}$, $\forall \boldsymbol{y}_k \in \mathbb{T}_f$ such that the successor state error $\boldsymbol{y}_{k+1} = f_{RK}(\hat{\boldsymbol{x}}_k, \kappa_f(\boldsymbol{y}_k)) - \hat{\boldsymbol{x}}_{k+1}^r \in \mathbb{T}_f$, $\forall \boldsymbol{y}_k \in \mathbb{T}_f$ and*

$$l_T(\boldsymbol{y}_{k+1}) + l(\boldsymbol{y}_k, \kappa_f(\boldsymbol{y}_k)) \leq l_T(\boldsymbol{y}_k)$$

which implies that $\mathbb{T}_f$ is a positive invariant set under $\kappa_f(\cdot)$, and $l_T(\cdot)$ is a control Lyapunov function in a neighbourhood of the origin.

By Theorem 1, the system state of (16) converges to zero. Therefore, by Theorem 2, the nominal system state $\hat{\boldsymbol{x}}$ can be regulated to its desired value $\hat{\boldsymbol{x}}_{N|k}^r$ within N steps if the prediction horizon N is sufficiently long [40]. Thus, inspired by [41], there exists a matrix K_f such that $A_K = A + BK_f$ is Hurwitz, where $A = (\partial h(\hat{\boldsymbol{x}}, \hat{\boldsymbol{u}})/\partial \hat{\boldsymbol{x}})|_{(\hat{\boldsymbol{x}}_{N|k}^r, \boldsymbol{0})}$ and $B = (\partial h(\hat{\boldsymbol{x}}, \hat{\boldsymbol{u}})/\partial \hat{\boldsymbol{u}})|_{(\hat{\boldsymbol{x}}_{N|k}^r, \boldsymbol{0})}$ with $h(\cdot)$ representing the nominal dynamics (53). Then, a locally stabilizing controller can be defined as $\kappa_f(\boldsymbol{y}_k) = K_f \boldsymbol{y}_k$, where K_f and Q_T can be determined by the following Lyapunov equation:

$$(A_K + \rho I)^\top Q_T + Q_T^\top (A_K + \rho I) + K_f^\top R K_f = -Q \tag{74}$$

where $\rho \in (0, \infty)$ satisfies $\rho < -\lambda_M(A_K)$. Moreover, there exists a constant $\beta > 0$ specifying an ellipsoidal terminal constraint $\mathbb{T}_f$ including the origin in the form of $\mathbb{T}_f := \{\|\boldsymbol{y}_k\|_{Q_T}^2 \leq \beta^2\}$ such that $\kappa_f(\boldsymbol{y}_k) \in \hat{\mathbb{Z}}$ holds for all $\boldsymbol{y}_k \in \mathbb{T}_f$. By Definition 1, $\mathbb{T}_f$ is positive invariant for the local control input $\kappa_f(\boldsymbol{y}_k)$, which is crucial to ensure recursive feasibility.

Remark 6. *Assumption 1 is standard in MPC stability analysis [23], [24]. The design philosophy for the controller $\kappa_f(\boldsymbol{y}_k)$ and the terminal constraint $\mathbb{T}_f$ follows from the result in [41], [42]. For more details, we refer the reader to Lemma 1 of [41].*

We are now ready to present the following proposition.

Proposition 1. *Suppose that Assumption 1 holds and the optimal control problem* (71) *is feasible at the initial time. Then, the optimization problem* (71) *is recursively feasible, and output regulation of the nominal system* (53) *is achieved, i.e., the tracking error dynamics* (70) *are asymptotically stable.*

The proof of Proposition 1 is provided in Appendix VIII-E.

Next, the solution to Problem 2 under the state and input constraints is obtained as follows.

Theorem 3. *Under Assumption 1, the constrained agile trajectory tracking and disturbance rejection Problem 2 for the uncertain quadrotor system composed of* (1)-(4) *can be solved by the following comprehensive control strategy:*

$$\begin{aligned} \boldsymbol{\vartheta}_{k+1} &= \begin{bmatrix} \boldsymbol{\vartheta}_{f,k+1} \\ \boldsymbol{\vartheta}_{\tau,k+1} \end{bmatrix} \\ &= \begin{bmatrix} M_f \boldsymbol{\vartheta}_{f,k} + N_f \boldsymbol{u}_{f,k} - M_f N_f \boldsymbol{v}_{e,k} \\ M_\tau \boldsymbol{\vartheta}_{\tau,k} + N_\tau \boldsymbol{u}_{\tau,k} - M_\tau N_\tau \bar{L}_0(\boldsymbol{\Omega}_{e,k}) \end{bmatrix} \end{aligned} \tag{75a}$$

$$\begin{aligned} \boldsymbol{u}_k^* &= \begin{bmatrix} f_{T,k}^* \\ \boldsymbol{\tau}_k^* \end{bmatrix} \\ &= \begin{bmatrix} (\hat{\boldsymbol{u}}_{f,0|k}^* + \Psi_f T_f^{-1} \boldsymbol{\vartheta}_{f,k} + \boldsymbol{\delta}_{f,k}(\boldsymbol{e}_{1,k}^*, \boldsymbol{e}_{2,k}^*)) \boldsymbol{z}_B \\ \hat{\boldsymbol{u}}_{\tau,0|k}^* + \Psi_\tau T_\tau^{-1} \boldsymbol{\vartheta}_{\tau,k} + \boldsymbol{\delta}_{\tau,k}(\tilde{\boldsymbol{e}}_{3,k}^*, \tilde{\boldsymbol{e}}_{4,k}^*, \boldsymbol{z}_k) \end{bmatrix} \end{aligned} \tag{75b}$$

where $\hat{\boldsymbol{u}}_{i,0|k}^*$ is defined in (73)*, $\Psi_i T_i^{-1}$ is defined in* (25)*, $\boldsymbol{\delta}_{i,k}$ is defined in* (60) *with $i \in \{f, \tau\}$, $\boldsymbol{x}_{e,k}^*$ is defined in* (58)*, and $\boldsymbol{z}_k$ is defined in* (44)*. The translational and rotational internal model compensators are defined in* (75a)*, and the nominal system-based NMPC optimization stabilizer is defined in* (75b)*.*

The proof of Theorem 3 is provided in Appendix VIII-F.

The roadmap of theoretical analysis is provided in Fig. 6.

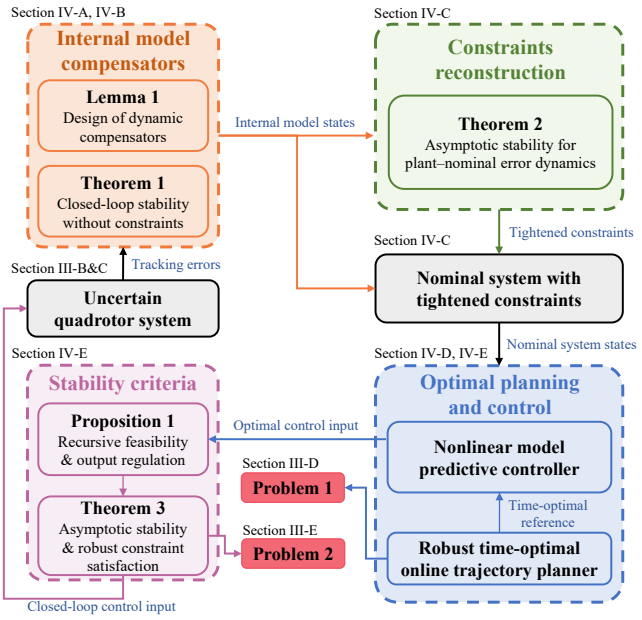


Fig. 6: Roadmap of theoretical analysis for the proposed framework.

V. Simulation Experiments

To demonstrate the effectiveness of the proposed framework, we design a set of experiments to address the following questions: (i) How does our time-optimal online planner perform in comparison to CPC [6], Sampling-based [30], Euclidean-based [19], TOGT [34], and PMM$_{cr}$ [21]? (ii) How does $\mathcal{IM}$-NMPO perform relative to the benchmark NMPC, INDI-NMPC [27], Online replanning MPCC (OR-MPCC) [21], and L_1 adaptive controller [5] based MPCC (L_1-MPCC) under parametric uncertainty and external disturbances?

To ensure a fair comparison, all benchmark algorithms are implemented and evaluated under identical conditions, including the quadrotor model, sampling and feedback rates, state estimation pipeline, model uncertainty, and wind disturbances. Except for our method and L_1-MPCC, the baseline controllers incorporate a linear drag model [36] to capture major aerodynamic effects. To further validate practical applicability, we deploy the proposed framework on a standard quadrotor with a 450 mm wheelbase, which is applied in the 2024 DJI Robomaster Intelligent MAV Championship Series.

To improve simulation realism, computational and communication delays are modeled by first-order low-pass filters, introducing phase lag to emulate real-world processing latency. Moreover, a time-adaptive reference strategy is applied to all controllers to compensate for delay-induced state mismatches. All optimization problems are formulated by CasADi [43] and numerically solved with the interior point optimizer (IPOPT) solver. The quadrotor parameters are listed in Table I.

A. Evaluation for Performance of Time-optimal Planner

This section evaluates the performance of the proposed polynomial-based time-optimal planner in two separate modes: the nominal dynamics model (53) with tightened constraints (64) (Ours$_{tc}$), and the original dynamics model (1)-(4) with the original constraints (6) (Ours$_{oc}$). We compare them to four baseline approaches - CPC [6], Sampling-based [30], Euclidean-based [19], and TOGT [34] - in terms of computational time (c.t.) and lap time (T_L). To ensure a fair comparison, all these methods are executed in a single global planning pass, as they do not support online replanning.

TABLE I: Quadrotor configurations

Parameters		Values
m	[kg]	1.00
l	[m]	0.45
J	[gm^2]	diag$(J_{11}, 0.0080, 0.0143)$
$(\boldsymbol{u}_{\min}, \boldsymbol{u}_{\max})$	[N]	(0.0, 4.18)
k_τ	[1]	0.0556

Table II and Fig. 7 show a clear trade-off between computational efficiency and trajectory optimality. CPC achieves competitive lap times but requires prohibitively long computation (e.g., over 30 minutes for the Race track). TOGT is the fastest in computation, but its lap times are consistently longer than those of our method. In contrast, the proposed planner strikes an effective balance, achieving computation times several orders of magnitude faster than CPC while consistently delivering shorter lap times than all baselines.

The advantage of TOGT in computation comes from its polynomial and differential-flatness formulation, but the simplified representation leads to conservative thrust usage and reduced agility (Fig. 7 (B)). Our method instead solves a full nonlinear time-optimal problem on the disturbance-decoupled nominal system, trading modest extra computation for substantially better lap performance. Furthermore, unlike the Euclidean-based method which uses straight-line distances, our polynomial-based waypoint allocation accounts for gate orientations, leading to more rational waypoint assignment and higher-quality trajectories, as visualized in Fig. 7 (A). Notably, the tightened-constraint variant Ours$_{tc}$ consistently achieves lower computation times than Ours$_{oc}$. In terms of lap times, Ours$_{oc}$ yields slightly shorter times on simpler, shorter tracks (Loop, Eight), while Ours$_{tc}$ attains better lap times on longer, more complex tracks (Split-S, Race). This performance trend is supported by the thrust profiles in Fig. 7 (B). As observed, although constraint tightening reduces the maximum available thrust, Ours$_{tc}$ operates more consistently near its thrust limit (average thrust 99.26% vs. 98.10% for Ours$_{oc}$), enabling it to better sustain high speeds over extended distances.

To further assess performance under online settings, we compare against the real-time replanning method PMM$_{cr}$ [21]. As shown in Table III, PMM$_{cr}$ achieves a marginally lower computation time (0.067 s vs. 0.099 s), yet Ours$_{tc}$ yields a shorter lap time (11.99 s vs. 12.56 s) and a higher maximum velocity (18.96 m/s vs. 18.31 m/s). The slightly longer computation of Ours$_{tc}$ is attributed to solving a full nonlinear time-optimal problem that respects the complete quadrotor dynamics, whereas PMM$_{cr}$ employs a simplified point-mass model. Fig. 8 further illustrates the replanning capability of our approach, where the central waypoint continuously moves.

B. Tracking Robustness Study

Upon obtaining the time-optimal trajectory, we compare the robustness of $\mathcal{IM}$-NMPO with the baseline NMPC, MPCC

TABLE II: Comparison of computation times and lap times

	CPC [6]		**Sampling** [30]		**Euclidean** [19]		**TOGT** [34]		**Ours**$_{oc}$		**Ours**$_{tc}$	
waypoints	c.t.[s]	T_L[s]	c.t.[s]	T_L[s]	c.t.[s]	T_L[s]	c.t.[s]	T_L[s]	c.t.[s]	T_L[s]	c.t.[s]	T_L[s]
Loop (4)	322.6	2.84	21.4	3.51	0.94	2.82	0.05	3.09	0.43	**2.81**	0.09	2.82
Eight (6)	374.7	5.13	36.1	5.99	2.11	5.15	0.29	5.93	0.89	**5.10**	0.46	5.13
Split-S (7)	1143	6.23	39.7	7.37	2.87	6.21	0.34	7.41	1.82	6.21	0.71	**6.16**
Race (8)	1978	12.18	73.1	13.46	4.17	12.23	1.13	13.23	3.31	12.16	1.89	**12.07**

It is noted that for a fair comparison, all these methods perform a single global planning.

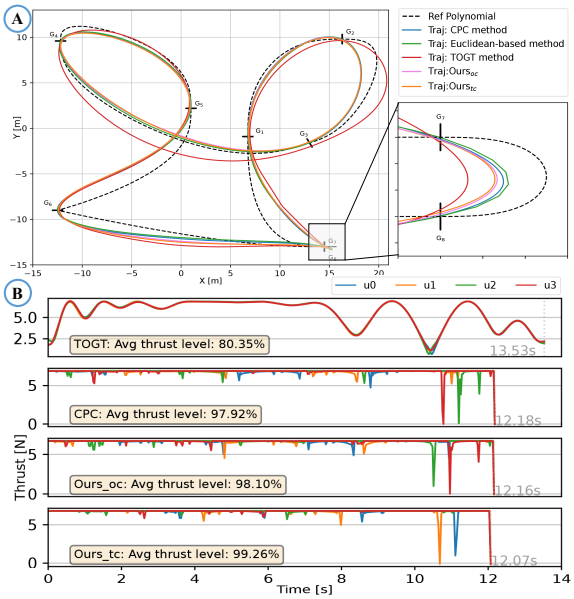


Fig. 7: (A) Simulation comparison of trajectories obtained from different planners. Note that the gate in the lower right corner consists of two gates stacked vertically. (B) Comparison of single-rotor thrust trajectories from TOGT, CPC, Ours$_{oc}$, and Ours$_{tc}$.

TABLE III: Comparison with the online time-optimal planner

Method	Lap time [s]	c.time [s]	Max vel [m/s]
PMM$_{cr}$	12.56	**0.067**	18.31
Ours$_{tc}$	**11.99**	0.099	**18.96**

[31], OR-MPCC [21], INDI-NMPC [27], and L_1-MPCC [5]. This study focuses on two critical challenges in agile flight.

1) Parametric uncertainty. The inertia matrix of the quadrotor contains some unknown parameters.
2) External disturbances. Constant, periodic, and stochastic torque disturbances are injected along $\boldsymbol{x}_B, \boldsymbol{y}_B$, and $\boldsymbol{z}_B$.

We quantitatively compare the tracking performance using the root mean square error (RMSE), cumulative RMSE (CRMSE), maximum position error (MaxErr), lap time (T_L), and success rate (SR). A trial is considered successful if the quadrotor sequentially passes through all gates while operating at its motion limits.

We first consider the impact of model uncertainty. The inertia matrix $J = \mathrm{diag}(J_{11}, 0.0080, 0.0143)$ contains an unknown parameter $\boldsymbol{\gamma} = J_{11}$. Through the preceding linearization process (18), we can calculate

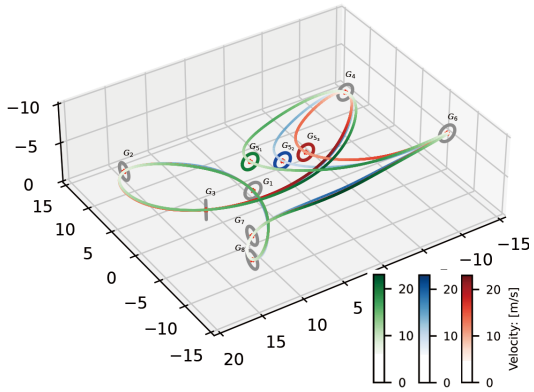


Fig. 8: Simulation involving moving waypoints. The green, blue and red gates represent the positions of the fifth gate at different times.

$$L_0(\boldsymbol{\beta}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.0080 & 0 \\ 0 & 0 & 0.0143 \end{bmatrix} \boldsymbol{\beta}, \; L_1(\boldsymbol{\beta}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \boldsymbol{\beta},$$

and hence $\bar{\mathcal{F}}_i(\boldsymbol{\Omega}_e, \boldsymbol{q}_e)$ and $\mathcal{H}_i(\boldsymbol{\Omega}_e, \boldsymbol{q}_e), i = 0, 1$ from (37). Then, the adaptive parameter update law is defined as

$$\dot{\hat{\boldsymbol{\gamma}}} = \Lambda_1 \left(\bar{\mathcal{F}}_1^{\top}(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) + \mathcal{H}_1^{\top}(\boldsymbol{\Omega}_e, \boldsymbol{q}_e) \right) \boldsymbol{\Omega}_e. \tag{76}$$

Table IV compares the tracking robustness under inertia uncertainty. With an accurate model, all methods achieve 100% success rates. NMPC gives the shortest lap time, while $\mathcal{IM}$-NMPO introduces a small robustness–performance trade-off. Meanwhile, $\mathcal{IM}$-NMPO achieves lower RMSE than NMPC and MPCC, and performs comparably to INDI-NMPC and L_1-MPCC, indicating its inherent capability to handle rotational aerodynamic effects. As the inertia uncertainty increases, NMPC and MPCC show clear degradation in RMSE, success rate, and lap time. INDI-NMPC and L_1-MPCC improve robustness, but their performance varies across different uncertainty levels. In contrast, $\mathcal{IM}$-NMPO consistently achieves the lowest RMSE, maintains high success rates, and keeps lap times close to the nominal case. This indicates that the proposed controller can effectively improve the robustness against model uncertainty.

We further evaluate the robustness against external torque disturbances using three disturbance categories: constant, periodic, and stochastic disturbances. The constant case applies a 1 Nm torque disturbance. For the periodic cases, the disturbance is defined as $d_i(t) = a_i \sin(\sigma_i t + \eta_i), \quad i = 1, 2, 3$ with two frequency settings: a low-frequency case (LF) ($\boldsymbol{\sigma} = [0.1, 0.2, 0.2]$) and a high-frequency case (HF) ($\boldsymbol{\sigma} = [15, 15, 16]$). The HF setting matches the dominant rotational frequency range observed in real flight data.

TABLE IV: Tracking performance comparisons of different methods under parametric uncertainty in the inertia matrix

	NMPC			INDI-NMPC			MPCC			L1-MPCC			$\mathcal{IM}$-NMPO		
	RMSE [m]	SR [%]	T_L [s]	RMSE [m]	SR [%]	T_L [s]	RMSE [m]	SR [%]	T_L [s]	RMSE [m]	SR [%]	T_L [s]	RMSE [m]	SR [%]	T_L [s]
Baseline	0.054	100	**12.21**	**0.048**	100	12.23	0.053	100	12.28	0.049	100	12.25	0.049	100	12.23
0.1 J_{11}	0.107	78.3	15.73	0.094	86.7	13.17	0.122	75.0	16.04	0.071	**96.7**	12.38	**0.054**	**96.7**	**12.31**
0.5 J_{11}	0.086	88.3	13.49	0.071	93.3	12.32	0.078	83.3	12.75	0.063	**100**	12.30	**0.050**	**100**	**12.26**
0.8 J_{11}	0.069	96.7	12.27	0.056	**100**	12.28	0.065	**100**	12.36	0.050	**100**	12.27	**0.049**	**100**	**12.23**
1.1 J_{11}	0.061	93.3	12.71	**0.049**	**100**	**12.22**	0.058	96.7	12.53	0.053	**100**	12.25	**0.049**	**100**	12.23
1.2 J_{11}	0.064	96.7	12.94	0.054	**100**	12.28	0.077	93.3	12.68	0.057	96.7	12.31	**0.049**	**100**	**12.23**
1.5 J_{11}	0.143	66.7	16.27	0.060	**96.7**	**12.39**	0.091	81.7	13.57	0.086	88.3	12.72	**0.058**	**96.7**	**12.39**

For each controller and parametric uncertainty condition, the RMSE, SR, and T_L are computed over 60 trials.

TABLE V: Tracking performance under various types of external torque disturbances

Method	0 Nm		1 Nm		LF periodic		HF periodic		OU		Pink noise	
	RMSE [m]	MaxErr [m]	RMSE [m]	MaxErr [m]	RMSE [m]	MaxErr [m]	RMSE [m]	MaxErr [m]	RMSE [m]	MaxErr [m]	RMSE [m]	MaxErr [m]
NMPC	0.054	0.16	–	–	1.28	3.41	0.28	0.69	0.32	0.84	0.39	1.11
OR-MPCC	0.050	0.15	–	–	0.69	2.34	0.26	0.62	0.30	0.78	0.35	0.96
L_1-ORMPCC	**0.046**	0.13	0.27	0.67	0.43	1.51	0.21	0.49	0.23	0.45	**0.17**	0.43
INDI-NMPC	0.048	0.14	0.23	0.87	0.59	1.75	0.19	0.42	**0.18**	0.43	0.18	0.44
$\mathcal{IM}$-NMPO	0.049	**0.10**	**0.13**	**0.34**	**0.24**	**0.47**	**0.17**	**0.32**	0.20	**0.41**	0.18	**0.40**

A dash '–' indicates that the controller failed to track the trajectory.

Then, we obtain $\Phi_{\tau,i} = \begin{bmatrix} 0 & 1 \\ -\sigma_i^2 & 0 \end{bmatrix}$, $\Psi_{\tau,i} = [1\ 0]$. Letting $M_{\tau,i} = \begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix}$, $N_{\tau,i} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and solving the Sylvester equation gives $T_{\tau,i}^{-1} = \begin{bmatrix} 3-\sigma_i^2 & 2 \\ -2\sigma_i^2 & 3-\sigma_i^2 \end{bmatrix}$. The unknown amplitudes and phases are $a_1 = 1$ Nm, $a_2 = 2$ Nm, $a_3 = 0.6$ Nm, $\eta_1 = 0.1$, $\eta_2 = 0.2$, and $\eta_3 = 0.3$.

For non-periodic stochastic cases, Ornstein–Uhlenbeck (OU) and pink-noise disturbances are introduced. For the OU case, each torque component is generated by

$$d\xi_i(t) = -\lambda_i \xi_i(t)dt + \rho_i dW_i(t), \quad d_i^{\mathrm{OU}}(t) = c_i \xi_i(t)$$

where $W_i(t)$ is a standard Wiener process, with $\boldsymbol{\lambda} = [1.0, 1.2, 1.4]$ and $\boldsymbol{\rho} = [1.0, 1.1, 0.6]$. Pink noise is generated by filtering white noise such that $S_{d_i}^{\mathrm{pink}}(f) \propto \frac{1}{f}$. Since neither disturbance has deterministic frequencies, a sliding-window FFT is applied to recent rotational tracking-error signals to estimate an effective local dominant frequency $\hat{\sigma}_i$.

The quantitative results are summarized in Table V. Under constant torque disturbance, NMPC and OR-MPCC fail to track the reference, while INDI-NMPC and L_1-ORMPCC suffer substantial RMSE increases compared with their nominal performance. In contrast, $\mathcal{IM}$-NMPO limits the degradation and achieves an RMSE of 0.13 m. Under the LF periodic disturbance, $\mathcal{IM}$-NMPO reduces the RMSE by 81.3% compared with NMPC and by at least 44.2% versus other robust baselines. For the HF periodic case, it also achieves the lowest RMSE/MaxErr of 0.17/0.32 m. These results show that the nonlinear internal models can actively learn and compensate for structured disturbances over a broad frequency range, as reflected by the internal-model state convergence in Fig. 9.

For the OU disturbance, INDI-NMPC achieves the lowest RMSE of 0.18 m, while $\mathcal{IM}$-NMPO obtains 0.20 m but reduces MaxErr to 0.41 m. Under pink-noise disturbance, L_1-ORMPCC gives the lowest RMSE of 0.17 m, while $\mathcal{IM}$-NMPO obtains 0.18 m but again achieves the smallest MaxErr of 0.40 m. This result clarifies the capability and limitation of the proposed framework: exact rejection is guaranteed for the finite-dimensional disturbance class, while broadband stochastic disturbances are empirically suppressed through locally dominant spectral components.

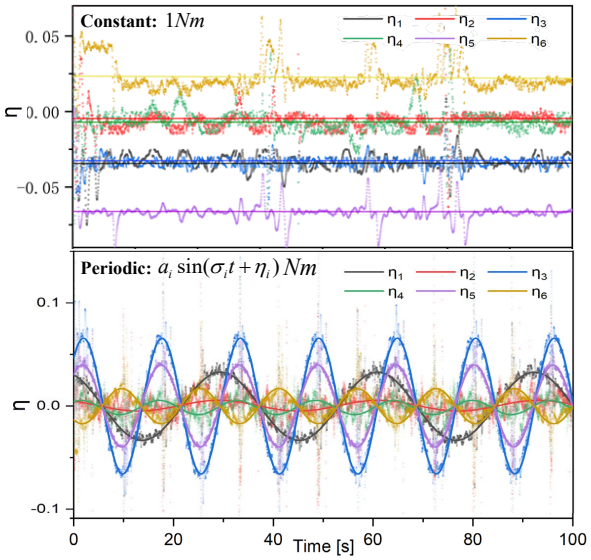


Fig. 9: The state of the internal models, converging to match the disturbance types (constant or periodic). Semi-transparent dots represent the raw sampled data and solid lines depict the fitted curves.

C. 2024 DJI Robomaster Intelligent MAV Championship

We first participated in the 2024 DJI Robomaster Intelligent MAV Championship for Planning and Control of Quadrotors, where the quadrotor was required to fly as fast as possible in a large-scale environment (120m × 100m × 45m) containing 12

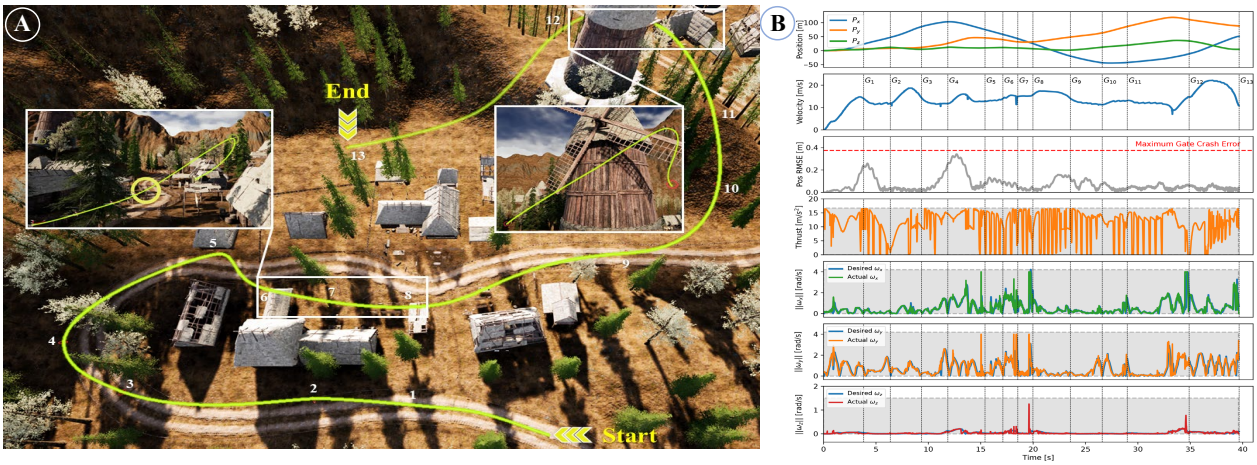


Fig. 10: 2024 DJI Robomaster Intelligent MAV Championship for Planning and Control Competition Results. (A) Bird's-eye view of the experiment environment. (B) Flight data of the Airsim experiment, including the position, velocity, tracking error, throttle, and angular velocity responses along the x, y, and z axes. The gray shaded regions denote the system constraint boundaries.

static gates and one dynamic pendulum gate (Fig. 10 (A)), under unknown persistent disturbances and battery voltage decay. The gate radius is 0.6 m and the quadrotor wheelbase is 450mm. Thus, the admissible corridor satisfies $\delta_j \leq 0.375$ m. To ensure clearance from the gate edges without excessive conservatism, we set $\delta_j = 0.25$ m.

With a receding horizon of $\phi = 3$ and an average replanning time of 97 ms, $\mathcal{IM}$-NMPO achieved a maximum speed of 23.3 m/s while keeping the tracking error below 0.375 m, as shown in Fig. 10 (B). The state and input constraints, including those on f_T and $\boldsymbol{\omega}$, are fully satisfied (i.e., the gray shaded regions). The vehicle maintains an average thrust acceleration of around 16.1 m/s^2, close to its physical limit (16.72 m/s^2). These results show that $\mathcal{IM}$-NMPO can safely operate near the quadrotor limits, while guaranteeing robust constraint satisfaction under unknown disturbances. The final rankings are reported in Table VI. Our developed system completed the racing track in 62 s, less than one-third of the time taken by the second-place. Readers can gain a better understanding of the experiment from the supplementary video.

TABLE VI: Ranking of Competition Results

rankings	**our team**	2nd place	3rd place	...
completion time	**62**s	190s	195s	...

Due to the competition limit of three attempts, a conservative strategy with 80% thrust limit was used to ensure a 100% success rate. While operating at full thrust, we can complete the course within **40** seconds.

To evaluate generalizability and robustness, we further raced in the 2024 DJI Robomaster Intelligent MAV Championship for Autonomous Perception and Racing of Quadrotors. In this competition, gate positions are provided before each race with up to 2 m uncertainty in each direction, requiring onboard visual detection and real-time replanning. We integrate $\mathcal{IM}$-NMPO with state estimation and vision-based gate detection. As shown in Fig. 11, the proposed method can rapidly replan time-optimal trajectories based on real-time visual detection results, completing the course in 72 s with speeds up to 10.9 m/s, improving upon the previous best record of 74 s.

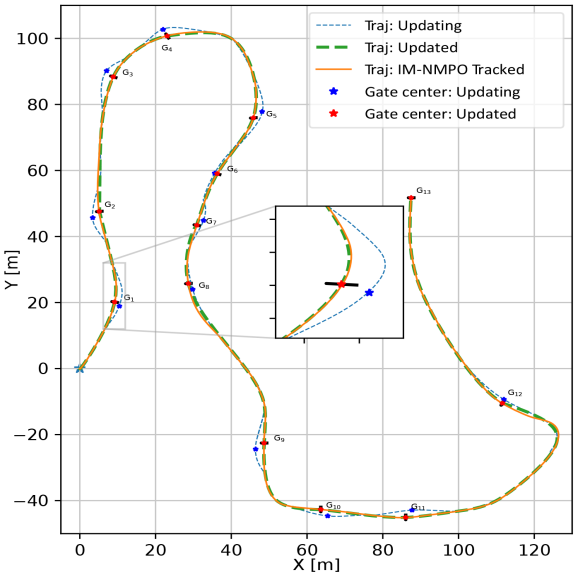


Fig. 11: Replanning and tracking performance of our strategy. Blue stars and dashed lines depict the real-time updated gate positions and receding-horizon time-optimal trajectories, respectively; red stars mark the final updated gate positions.

VI. Real-World Experiments

To further validate the robustness and generalizability of the proposed framework, we conduct agile autonomous flight tests in various real-world environments, including a compact indoor laboratory ($11\times6\times4$m^3), an outdoor park, and a cluttered dynamic environment ($26\times9\times9$m^3), as shown in Fig. 1. The experiments are performed on two quadrotors with different wheelbases (250 mm and 330 mm) and thrust-to-weight ratios (3.0 and 2.1). An Intel NUC13 onboard computer performs all real-time computation, communicating with a Pixhawk5 flight controller running PX4 firmware for low-level control. State estimation is provided by the PX4 extended Kalman filter (EKF), fusing inertial measurement unit data with position and orientation from an OptiTrack motion-capture system, a global positioning system (GPS) or an onboard vision sensor.

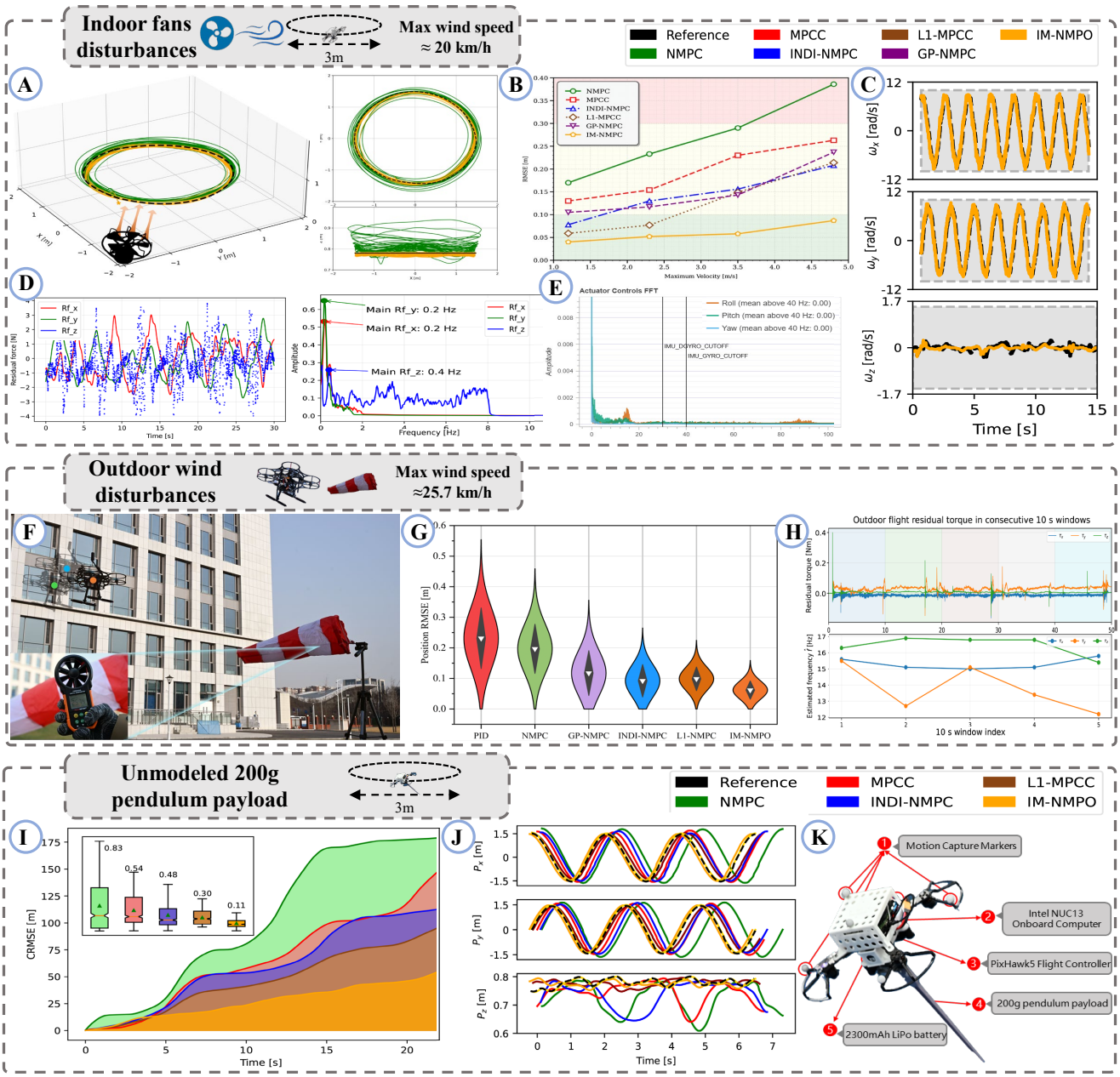


Fig. 12: Trajectory tracking results with unknown disturbances and model uncertainty. (A) Top and side view of the indoor fan-disturbance tracking performance. (B) Position tracking errors as a function of maximum velocity. (C) Angular velocity responses along x, y, and z axes. (D-E) Finite-window FFT analysis of indoor flight. (F) Hovering test in the outdoor park. (G) Violin plots of position RMSE under wind disturbances. (H) Finite-window FFT analysis of residual-torque signals over consecutive 10 s windows. (I) CRMSE under model mismatch. (Inset) Average RMSE. (J) Position tracking comparison. (K) An unmodeled pendulum payload of 200g is added to the quadrotor.

A. *Unknown Indoor Fan Disturbances*

This experiment evaluates tracking accuracy under persistent fan-induced wind disturbances. The quadrotor tracks a circular trajectory with a radius of 1.5 m in a wind field covering the flight area, where the wind speed at the trajectory center reaches 20 km/h. The flight speed is gradually increased up to 4.8 m/s, and $\mathcal{IM}$-NMPO is compared with NMPC, MPCC, INDI-NMPC, L_1-MPCC, and GP-NMPC [9].

As shown in Fig. 12 (A), our method closely follows the reference trajectory, whereas NMPC exhibits considerable degradation. Fig. 12 (B) further shows that $\mathcal{IM}$-NMPO achieves the lowest tracking error across all tested speeds. At the highest speed, it reduces the position tracking error by 60% relative to INDI-NMPC, 58% relative to L_1-MPCC, and 63% relative to GP-NMPC.

The underlying mechanism for this robustness is further examined through spectral analysis. Finite-window FFT analysis of the flight data at the highest speed, using consecutive 10 s windows, shows locally dominant components around 0.2–0.4 Hz in the translational dynamics and around 15 Hz in the rotational dynamics (Fig. 12 (D) and (E)). These results indicate that our framework can learn and compensate for locally dominant disturbance components in real flight, improving tracking robustness under persistent wind disturbances.

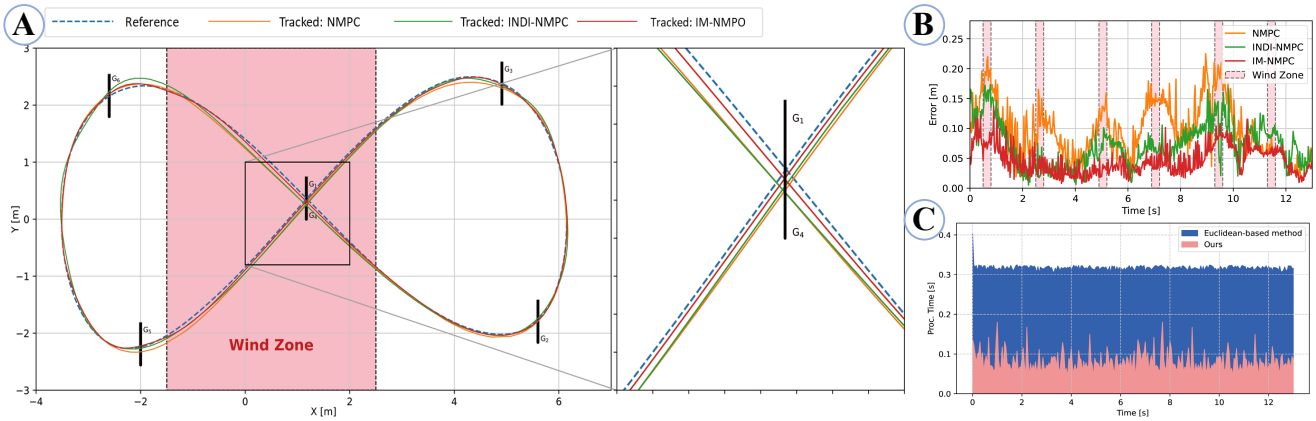


Fig. 13: Drone-laboratory racing under 35–40 km/h wind disturbances, with speeds up to 10 m/s. (A) Flight trajectories projected on the x-y plane; the shaded pink region denotes the wind zone. (B) Tracking errors over three flight loops. (C) CPU time comparison.

B. Unknown Outdoor Wind Disturbances

To further evaluate robustness under general disturbances, we conduct outdoor hovering experiments in moderate breeze conditions, with measured wind speeds up to 25.7 km/h. Unlike the indoor fan tests, outdoor wind contains broadband and time-varying components that are neither strictly sinusoidal nor repeatable. The quadrotor hovers at 3.5 m height, using only an onboard GPS receiver fused with the EKF, resulting in lower-precision state estimation than motion capture.

As shown in Fig. 12 (G), $\mathcal{IM}$-NMPO achieves a position RMSE of 0.062 m, outperforming INDI-NMPC by 35.1% and L_1-MPCC by 36.7%. These results show that our framework remains effective under non-periodic and unpredictable wind disturbances. Theoretically, the disturbance model (7) does not require real disturbances to be perfectly sinusoidal. Wind disturbances, even when stochastic, typically exhibit locally dominant frequency components over finite time windows. Fig. 12 (H) shows that outdoor residual-torque signals are divided into consecutive 10 s windows, and the estimated dominant frequencies vary across windows rather than form a stationary global harmonic signature. This suggests that the framework is not limited to strictly periodic disturbances and can effectively mitigate general disturbances in practice.

C. Unknown Model Mismatch

This experiment evaluates the capability to compensate for real-time parametric uncertainty in the inertia matrix J. A 200 g unmodeled pendulum payload, constrained to swing only along the $\boldsymbol{y}_B$ axis, is attached to the quadrotor (Fig. 12 (K)). This results in real-time variations in the I_{yy} component of J. All controllers are not pre-adapted to account for the changes in the inertia matrix and execute the same circular trajectory described in Section VI-A.

Fig. 12 (I) and (J) demonstrate that with the unknown pendulum load attached, $\mathcal{IM}$-NMPO achieves a noticeable reduction (69%) of the position RMSE compared to the baseline NMPC (0.07 m vs. 0.23 m). Furthermore, the proposed method outperforms INDI-NMPC by 46% and L_1-MPCC by 41%, indicating that $\mathcal{IM}$-NMPO can rapidly estimate and compensate for time-varying inertia uncertainty during flight.

D. Drone Laboratory Racing with Wind Zone

In this section, we validate the proposed framework in a compact indoor flight space ($11\times6\times4\text{m}^3$), where the quadrotor must navigate six waypoints in sequence (Fig. 1). Two industrial fans placed near the central gates create a strong wind zone of 35–40 km/h, measured by an anemometer.

As shown in Fig. 13, $\mathcal{IM}$-NMPO achieves a 50% improvement compared to NMPC and outperforms INDI-NMPC by less than 5 cm in RMSE. We achieve a maximum flight speed of 10 m/s in the compact indoor environment while maintaining tracking error below 0.14 m, even in the wind zone. Fig. 13 (C) shows the CPU time of our planner and the Euclidean-based method [19]. The computation time of our method remains below 0.10 s, which is significantly lower than the Euclidean-based method (0.32 s). These results underscore the framework's ability to ensure accurate, robust, and computationally efficient flight under unpredictable disturbances.

E. Safe Agile Navigation in Cluttered Dynamic Environments

This experiment aims to verify that the proposed framework enables the quadrotor to autonomously fly through cluttered dynamic environments without motion capture systems. This environment features eight static gates, two static pillars, two dynamic pillars, a perforated obstacle, and a high-density obstacle zone generated by three industrial fans with wind speeds of 10-25 km/h (Fig. 14 (A)). Using only onboard sensing and computation, the quadrotor must sequentially pass through all gates, avoid static/dynamic obstacles and navigate the wind zone, starting and ending at the same point $\boldsymbol{G}_0$.

We employ VINS-Fusion [44] for visual-inertial odometry and YOLOv8n-Pose [45] for gate and obstacle detection. As shown in Fig. 14 (B), when entering the wind zone, $\mathcal{IM}$-NMPO rapidly compensates for the crosswind and restores the tracking error to its original level (less than 0.1 m) within two seconds. The maximum tracking error remains below 0.19 m, and the three-axis angular velocity responses tightly track their respective desired values. This experiment was successfully repeated multiple times, providing empirical support for the practical robustness of the proposed framework in cluttered dynamic environments with unknown disturbances and the additional uncertainty introduced by onboard vision sensing.

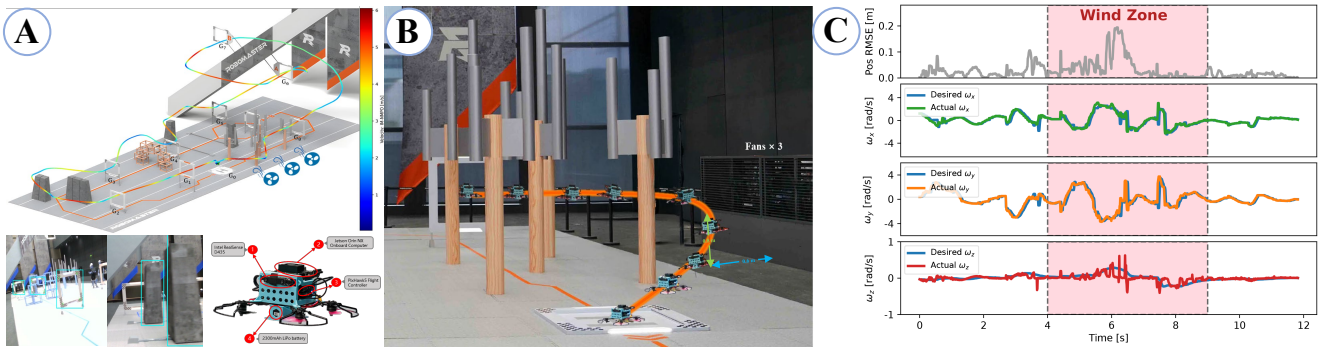


Fig. 14: Safe agile navigation in cluttered environments. (A) Overview of the experiment environment and the vision-based quadrotor vehicle. (B) Snapshots of the quadrotor traversing the wind zone, with the closest distance to the fans being 0.6 m. (C) Flight data in the wind zone.

VII. Conclusion

This paper proposes $\mathcal{IM}$-NMPO, an internal model-based nonlinear model predictive optimization framework for robust optimal planning and control of agile quadrotor flight under unknown payloads, persistent winds, and time-varying gusts.

The core insight is the systematic decoupling of uncertain dynamics from the optimization loop using the IMP, allowing both time-optimal planning and NMPC to operate on the disturbance-decoupled nominal system. The proposed framework integrates nonlinear internal model compensators for real-time disturbance rejection and a nominal system-based NMPC stabilizer for high-accuracy, constrained agile trajectory tracking. A formal technical analysis is performed to prove that the controlled quadrotor achieves robust constraint satisfaction, recursive feasibility, and asymptotic stability. A receding-horizon polynomial waypoint-allocation module is further proposed for efficient online time-optimal reference generation. Extensive experiments under various disturbances and quadrotor configurations demonstrate that our framework can effectively address the challenges of robust agile quadrotor flight under unknown disturbances and parametric uncertainty.

$\mathcal{IM}$-NMPO advances and offers insights into both **output regulation** and **nonlinear model predictive optimization**, providing a unified framework for agile and robust motion against multiple uncertainties of real-world robotic systems.

VIII. Appendix

A. Proof of Lemma 1

Since the augmented system state $\tilde{\boldsymbol{x}}$ is bounded, by (48), $\dot{\tilde{\boldsymbol{x}}}$ is bounded. By (14) and (35), $\boldsymbol{q}_e$ and $\boldsymbol{\omega}_e$ are also bounded. By (33), (42), (45) and the boundedness of $\boldsymbol{v}_e, \boldsymbol{\Omega}_e, \boldsymbol{z}, \tilde{\boldsymbol{\vartheta}}_f$, and $\tilde{\boldsymbol{\vartheta}}_\tau$, one has that $\bar{\boldsymbol{\vartheta}}_f$, $\bar{\boldsymbol{\vartheta}}_\tau$, and $\dot{\tilde{\boldsymbol{\vartheta}}}_\tau$ are bounded. Then, by (25), (30) and (38), $\boldsymbol{\vartheta}_f$ and $\boldsymbol{\vartheta}_\tau$ are bounded. Furthermore, by (12), the boundedness of $\boldsymbol{x}^r$ implies the boundedness of $\boldsymbol{x}$. Given that $\lim_{t\to\infty}(\boldsymbol{p}_e(t), \boldsymbol{v}_e(t), \tilde{\boldsymbol{q}}_e(t), \boldsymbol{\Omega}_e(t)) = (\boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0})$, Problem 2 can be solved by the controller composed of (51) and (52).

B. Proof of Theorem 1

Since M_i is Hurwitz, there exists a symmetric positive definite matrix L_i such that $L_i M_i + M_i^\top L_i = -I$, $i \in \{f, \tau\}$. Define the parameter estimation error $\tilde{\boldsymbol{\gamma}} = \hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}$ and the positive definite function $V(t) = V_1(t) + V_2(t)$, where

$$\begin{aligned} V_1(t) &= \frac{1}{2} k_p \boldsymbol{p}_e^\top \boldsymbol{p}_e + \epsilon_f \tilde{\boldsymbol{\vartheta}}_f^\top L_f \tilde{\boldsymbol{\vartheta}}_f + \frac{1}{2} \boldsymbol{v}_e^\top \boldsymbol{v}_e \\ V_2(t) &= k_q \tilde{\boldsymbol{q}}_e^\top \tilde{\boldsymbol{q}}_e + \epsilon_\tau \tilde{\boldsymbol{\vartheta}}_\tau^\top L_\tau \tilde{\boldsymbol{\vartheta}}_\tau + \frac{1}{2} \boldsymbol{\Omega}_e^\top J \boldsymbol{\Omega}_e + \frac{1}{2} \tilde{\boldsymbol{\gamma}}^\top \Lambda_1^{-1} \tilde{\boldsymbol{\gamma}} \end{aligned} \tag{77}$$

with positive real numbers ϵ_f and ϵ_τ to be determined.

First, the time derivative of $V_1(t)$ along the trajectory of the translational dynamic subsystem in (48) satisfies

$$\begin{aligned} \dot{V}_1(t) =& k_p \dot{\boldsymbol{p}}_e^\top \boldsymbol{p}_e + \epsilon_f \dot{\tilde{\boldsymbol{\vartheta}}}_f^\top L_f \tilde{\boldsymbol{\vartheta}}_f + \epsilon_f \tilde{\boldsymbol{\vartheta}}_f^\top L_f \dot{\tilde{\boldsymbol{\vartheta}}}_f + \boldsymbol{v}_e^\top \dot{\boldsymbol{v}}_e \\ =& -\epsilon_f \|\tilde{\boldsymbol{\vartheta}}_f\|^2 + \boldsymbol{v}_e^\top P_f \tilde{\boldsymbol{\vartheta}}_f - \boldsymbol{v}_e^\top K_v \boldsymbol{v}_e \\ \le& -\frac{1}{2}\epsilon_f \|\tilde{\boldsymbol{\vartheta}}_f\|^2 - (\lambda_m(K_v) - \frac{1}{2\epsilon_f}\|P_f\|^2)\|\boldsymbol{v}_e\|^2 \\ \le& -\frac{1}{2}\epsilon_f \|\tilde{\boldsymbol{\vartheta}}_f\|^2 - \|\boldsymbol{v}_e\|^2 \end{aligned} \tag{78}$$

where $P_f = \Psi_f T_f^{-1}$ and $\epsilon_f \ge \frac{1}{2(\lambda_m(K_v)-1)}\|P_f\|^2$.

Then, the time derivative of $V_2(t)$ along the trajectory of the rotational dynamic subsystem in (48) satisfies

$$\begin{aligned} \dot{V}_2(t) =& k_q \tilde{\boldsymbol{q}}_e^\top (\boldsymbol{q}_1 \odot q(\boldsymbol{\Omega}_e - E_1 \boldsymbol{q}_{ev})) - \epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 + \boldsymbol{\Omega}_e^\top (P_\tau \tilde{\boldsymbol{\vartheta}}_\tau \\ & - k_q \boldsymbol{q}_{ev} - K_\Omega \boldsymbol{\Omega}_e - \rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\tilde{\boldsymbol{\gamma}}) \\ & + \tilde{\boldsymbol{\gamma}}^\top \Lambda_1^{-1} \Lambda_1 \rho^\top(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z}) \boldsymbol{\Omega}_e \\ =& k_q \boldsymbol{q}_{ev}^\top \boldsymbol{\Omega}_e - k_q \boldsymbol{q}_{ev}^\top E_1 \boldsymbol{q}_{ev} - \epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 + \boldsymbol{\Omega}_e^\top P_\tau \tilde{\boldsymbol{\vartheta}}_\tau \\ & - k_q \boldsymbol{\Omega}_e^\top \boldsymbol{q}_{ev} - \boldsymbol{\Omega}_e^\top K_\Omega \boldsymbol{\Omega}_e - \boldsymbol{\Omega}_e^\top \rho(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z})\tilde{\boldsymbol{\gamma}} \\ & + \tilde{\boldsymbol{\gamma}}^\top \Lambda_1^{-1} \Lambda_1 \rho^\top(\boldsymbol{\Omega}_e, \boldsymbol{q}_e, \boldsymbol{z}) \boldsymbol{\Omega}_e \\ \le& -k_q \boldsymbol{q}_{ev}^\top E_1 \boldsymbol{q}_{ev} - \epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 + \left\|\boldsymbol{\Omega}_e^\top P_\tau\right\|^2 \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 \\ & - \boldsymbol{\Omega}_e^\top K_\Omega \boldsymbol{\Omega}_e \\ \le& -\frac{1}{2}\epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 - (\lambda_m(K_\Omega) - \frac{1}{2\epsilon_\tau}\|P_\tau\|^2)\|\boldsymbol{\Omega}_e\|^2 \\ & - k_q \lambda_m(E_1)\|\boldsymbol{q}_{ev}\|^2 \\ \le& -\frac{1}{2}\epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 - \|\boldsymbol{\Omega}_e\|^2 - k_q \lambda_m(E_1)\|\boldsymbol{q}_{ev}\|^2 \end{aligned} \tag{79}$$

where $P_\tau = \Psi_\tau T_\tau^{-1}$ and $\epsilon_\tau \ge \frac{1}{2(\lambda_m(K_\Omega)-1)}\|P_\tau\|^2$.

By (78) and (79), we have

$$\begin{aligned} \dot{V}(t) \le& -\frac{1}{2}\epsilon_f \|\tilde{\boldsymbol{\vartheta}}_f\|^2 - \frac{1}{2}\epsilon_\tau \|\tilde{\boldsymbol{\vartheta}}_\tau\|^2 - \|\boldsymbol{v}_e\|^2 - \|\boldsymbol{\Omega}_e\|^2 \\ & - k_q \lambda_m(E_1)\|\boldsymbol{q}_{ev}\|^2 . \end{aligned} \tag{80}$$

Since $V(t) \ge 0$, (80) implies the boundedness of $\boldsymbol{p}_e$, $\boldsymbol{v}_e$, $\tilde{\boldsymbol{\vartheta}}_f$, $\tilde{\boldsymbol{q}}_e$, $\boldsymbol{\Omega}_e$, $\tilde{\boldsymbol{\vartheta}}_\tau$, and $\tilde{\boldsymbol{\gamma}}$. It then follows that $\boldsymbol{q}_e$ and $\hat{\boldsymbol{\gamma}}$ are also bounded. Moreover, by (44), $\boldsymbol{z}$ is bounded. Thus, it can be concluded that the state $\tilde{\boldsymbol{x}}$ of the augmented system

is bounded. Furthermore, by (48) and (51), $\dot{\tilde{\boldsymbol{x}}}$ is bounded. Then, $\ddot{V}(t)$ is bounded. Therefore, by Barbalat's lemma [46], $\lim_{t\to\infty}\dot{V}(t)=0$, which means

$$\lim_{t\to\infty}(\tilde{\boldsymbol{\vartheta}}_f(t),\tilde{\boldsymbol{\vartheta}}_\tau(t),\boldsymbol{v}_e(t),\boldsymbol{\Omega}_e(t),\boldsymbol{q}_{ev}(t))=(\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0}).$$

By (14) and (35), we have that $\lim_{t\to\infty}\tilde{\boldsymbol{q}}_e(t)=\mathbf{0}$ and $\lim_{t\to\infty}\boldsymbol{\omega}_e(t)=\mathbf{0}$.

Furthermore, by (48) and (51), we have

$$\dot{\boldsymbol{v}}_e=-k_p\boldsymbol{p}_e-K_v\boldsymbol{v}_e+\Psi_f T_f^{-1}\tilde{\boldsymbol{\vartheta}}_f. \tag{81}$$

Then,

$$\ddot{\boldsymbol{v}}_e=-k_p\dot{\boldsymbol{p}}_e-K_v\dot{\boldsymbol{v}}_e+\Psi_f T_f^{-1}\dot{\tilde{\boldsymbol{\vartheta}}}_f. \tag{82}$$

Since $\dot{\tilde{\boldsymbol{x}}}$ is bounded, it follows that $\ddot{\boldsymbol{v}}_e$ is bounded. Again, by Barbalat's lemma [46], $\lim_{t\to\infty}\dot{\boldsymbol{v}}_e(t)=\mathbf{0}$, which, in view of (81), implies that $\lim_{t\to\infty}\boldsymbol{p}_e(t)=\mathbf{0}$. The proof is complete by invoking Lemma 1.

C. Proof of Lemma 2

Since $\boldsymbol{\varpi}(t)$ is bounded and converges to zero asymptotically, $g(t)\triangleq a\|\boldsymbol{\varpi}(t)\|^2$ is bounded and satisfies $\lim_{t\to\infty}g(t)=0$. For any given $\epsilon>0$, whenever $\|\boldsymbol{\mu}(t)\|^2>\frac{g(t)+\epsilon}{k}$, it follows from (61) that $\dot{W}(t)<-\epsilon$. This implies that the trajectories of $W(t)$ cannot grow unboundedly. Specifically, $W(t)$ is bounded for all $t\geq 0$, which directly guarantees the boundedness of $\boldsymbol{\mu}(t)$ and $\boldsymbol{\nu}(t)$. This completes the proof.

D. Proof of Theorem 2

Define the parameter estimation error $\tilde{\boldsymbol{\gamma}}=\hat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}$ and the positive definite function $V_e(t)=V_1(t)+V_2(t)$, where

$$\begin{aligned}V_1(t)&=\frac{1}{2}k_1\boldsymbol{e}_1^\top\boldsymbol{e}_1+\frac{1}{2}\boldsymbol{e}_2^\top\boldsymbol{e}_2\\ V_2(t)&=k_3\tilde{\boldsymbol{e}}_3^\top\tilde{\boldsymbol{e}}_3+\frac{1}{2}\tilde{\boldsymbol{e}}_4^\top J\tilde{\boldsymbol{e}}_4+\frac{1}{2}\tilde{\boldsymbol{\gamma}}^\top\Lambda_2^{-1}\tilde{\boldsymbol{\gamma}}.\end{aligned} \tag{83}$$

Denote

$$\begin{aligned}\boldsymbol{r}_f&:=\Psi_f T_f^{-1}\left(\tilde{\boldsymbol{\vartheta}}_f+N_f\boldsymbol{v}_e\right)\\ \boldsymbol{r}_\tau&:=\Psi_\tau T_\tau^{-1}\left(\tilde{\boldsymbol{\vartheta}}_\tau+N_\tau J\boldsymbol{\Omega}_e\right)\end{aligned} \tag{84}$$

First, the time derivative of $V_1(t)$ satisfies:

$$\begin{aligned}\dot{V}_1(t)=&k_1\boldsymbol{e}_2^\top\boldsymbol{e}_1+\boldsymbol{e}_2^\top(f_T\boldsymbol{z}_B+m^{-1}\boldsymbol{d}_f-\hat{\boldsymbol{u}}_f)\\ =&-\boldsymbol{e}_2^\top K_2\boldsymbol{e}_2+\boldsymbol{e}_2^\top\boldsymbol{r}_f\\ \leq&-\lambda_m(K_2)\|\boldsymbol{e}_2\|^2+\boldsymbol{e}_2^\top\boldsymbol{r}_f.\end{aligned} \tag{85}$$

Then, the time derivative of $V_2(t)$ satisfies:

$$\begin{aligned}\dot{V}_2(t)=&2k_3\tilde{\boldsymbol{e}}_3^\top\dot{\tilde{\boldsymbol{e}}}_3+\tilde{\boldsymbol{e}}_4^\top J\dot{\tilde{\boldsymbol{e}}}_4+\tilde{\boldsymbol{\gamma}}^\top\Lambda_2^{-1}\dot{\tilde{\boldsymbol{\gamma}}}\\ =&-k_3\tilde{\boldsymbol{e}}_{3v}^\top E_2\tilde{\boldsymbol{e}}_{3v}-\tilde{\boldsymbol{e}}_4^\top K_4\tilde{\boldsymbol{e}}_4+\tilde{\boldsymbol{e}}_4^\top\boldsymbol{r}_\tau\\ \leq&-k_3\lambda_m(E_2)\|\tilde{\boldsymbol{e}}_{3v}\|^2-\lambda_m(K_4)\|\tilde{\boldsymbol{e}}_4\|^2+\tilde{\boldsymbol{e}}_4^\top\boldsymbol{r}_\tau.\end{aligned} \tag{86}$$

Thus, by (85) and (86), we have

$$\begin{aligned}\dot{V}_e(t)=&\dot{V}_1(t)+\dot{V}_2(t)\\ \leq&-\lambda_m(K_2)\|\boldsymbol{e}_2\|^2-k_3\lambda_m(E_2)\|\tilde{\boldsymbol{e}}_{3v}\|^2-\lambda_m(K_4)\|\tilde{\boldsymbol{e}}_4\|^2\\ &+\boldsymbol{e}_2^\top\boldsymbol{r}_f+\tilde{\boldsymbol{e}}_4^\top\boldsymbol{r}_\tau\\ \leq&-\frac{1}{2}\lambda_m(K_2)\|\boldsymbol{e}_2\|^2-k_3\lambda_m(E_2)\|\tilde{\boldsymbol{e}}_{3v}\|^2-\frac{1}{2}\lambda_m(K_4)\|\tilde{\boldsymbol{e}}_4\|^2\\ &+\frac{1}{2\lambda_m(K_2)}\|\boldsymbol{r}_f\|^2+\frac{1}{2\lambda_m(K_4)}\|\boldsymbol{r}_\tau\|^2.\end{aligned} \tag{87}$$

Let $\boldsymbol{\mu}_e:=\mathrm{col}\,[e_2,\tilde{\boldsymbol{e}}_{3v},\tilde{\boldsymbol{e}}_4]$ and $\boldsymbol{\varpi}_e:=\mathrm{col}\,[\boldsymbol{r}_f,\boldsymbol{r}_\tau]$. From (87),

$$\dot{V}_e(t)\leq-k_e\|\boldsymbol{\mu}_e(t)\|^2+a_e\|\boldsymbol{\varpi}_e(t)\|^2 \tag{88}$$

where $k_e=\min\left\{\frac{1}{2}\lambda_m(K_2),k_3\lambda_m(E_2),\frac{1}{2}\lambda_m(K_4)\right\}>0$, and $a_e=\max\left\{\frac{1}{2\lambda_m(K_2)},\frac{1}{2\lambda_m(K_4)}\right\}>0$. By Theorem 1, $\tilde{\boldsymbol{\vartheta}}_f$, $\tilde{\boldsymbol{\vartheta}}_\tau$, $\boldsymbol{v}_e$, and $\boldsymbol{\Omega}_e$ are bounded and converge to the origin. Then, by (84), $\boldsymbol{r}_f$ and $\boldsymbol{r}_\tau$ are bounded and converge to the origin. Thus, $\boldsymbol{\varpi}_e(t)$ is bounded and satisfies $\lim_{t\to\infty}\boldsymbol{\varpi}_e(t)=\mathbf{0}$. By Lemma 2, $V_e(t)$ is bounded for all $t\geq 0$, which means the boundedness of $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\tilde{\boldsymbol{e}}_3$, $\tilde{\boldsymbol{e}}_4$, and $\tilde{\boldsymbol{\gamma}}$. It then follows that $\boldsymbol{e}_3$ and $\hat{\boldsymbol{\gamma}}$ are also bounded. Furthermore, by (58) and (60), it can be obtained that $\dot{\boldsymbol{x}}_e$ is bounded. Therefore, by the generalized Barbalat's lemma [47], $\lim_{t\to\infty}\dot{V}_e(t)=0$, which implies that

$$\lim_{t\to\infty}(\boldsymbol{e}_2(t),\tilde{\boldsymbol{e}}_{3v}(t),\tilde{\boldsymbol{e}}_4(t))=(\mathbf{0},\mathbf{0},\mathbf{0}).$$

By (57), we have $\lim_{t\to\infty}\tilde{\boldsymbol{e}}_3(t)=\mathbf{0}$ and $\lim_{t\to\infty}\boldsymbol{e}_4(t)=\mathbf{0}$.

Furthermore, by (58) and (60), we have

$$\dot{\boldsymbol{e}}_2=-k_1\boldsymbol{e}_1-K_2\boldsymbol{e}_2+\boldsymbol{r}_f. \tag{89}$$

Then,

$$\ddot{\boldsymbol{e}}_2=-k_1\dot{\boldsymbol{e}}_1-K_2\dot{\boldsymbol{e}}_2+\dot{\boldsymbol{r}}_f. \tag{90}$$

Since $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{r}_f$, $\tilde{\boldsymbol{\vartheta}}_f$, and $\boldsymbol{v}_e$ are bounded, by (48) and (84), we have $\dot{\boldsymbol{e}}_1$, $\dot{\boldsymbol{e}}_2$, and $\dot{\boldsymbol{r}}_f$ are bounded. Thus, $\ddot{\boldsymbol{e}}_2$ is bounded. Again, by Barbalat's lemma [46], $\lim_{t\to\infty}\dot{\boldsymbol{e}}_2(t)=\mathbf{0}$, which, in view of (89), implies that $\lim_{t\to\infty}\boldsymbol{e}_1(t)=\mathbf{0}$.

E. Proof of Proposition 1

For each integer k, let $\hat{\boldsymbol{u}}_k=\{\hat{\boldsymbol{u}}_{0|k},\ \hat{\boldsymbol{u}}_{1|k},\dots,\hat{\boldsymbol{u}}_{N-1|k}\}$ denote a feasible control sequence that steers the tracking error $\boldsymbol{y}_k$ into the terminal constraint $\mathbb{T}_f$ within N steps or less [40], with $\hat{\boldsymbol{u}}_{i|k}\in\hat{\mathbb{Z}}$ for $i=0,1,\dots,N-1$. The corresponding state sequence is denoted by $\hat{\boldsymbol{x}}_k=\{\hat{\boldsymbol{x}}_{0|k},\hat{\boldsymbol{x}}_{1|k},\dots,\hat{\boldsymbol{x}}_{N|k}\}$, with $\hat{\boldsymbol{x}}_{i|k}\in\hat{\mathbb{X}}$ and $\boldsymbol{y}_{N|k}\in\mathbb{T}_f$. Suppose then, the optimal control sequence $\hat{\boldsymbol{u}}_k^*$ and the optimal state trajectory $\hat{\boldsymbol{x}}_k^*$ in time step k have been determined in (72). Then, the control policy (73) steers the initial state $\hat{\boldsymbol{x}}_{0|k}^*$ to the successor state $\hat{\boldsymbol{x}}_{0|k+1}=\hat{\boldsymbol{x}}_{1|k}^*$. Our objective is to obtain an admissible control sequence $\hat{\boldsymbol{u}}_{k+1}$ for $\hat{\boldsymbol{x}}_{0|k+1}$ and hence, an upper bound for $J(\boldsymbol{y}_{k+1},\hat{\boldsymbol{u}}_{k+1})$ at time step $k+1$.

Define the candidate sequence by shifting the optimal inputs and appending the terminal law:

$$\begin{aligned}\hat{\boldsymbol{u}}_{i|k+1}^c&=\begin{cases}\hat{\boldsymbol{u}}_{i+1|k}^*, & i=0,\dots,N-2\\ \kappa_f(\hat{\boldsymbol{y}}_{N|k}^*), & i=N-1\end{cases}\\ \hat{\boldsymbol{x}}_{i|k+1}^c&=\hat{\boldsymbol{x}}_{i+1|k}^*,\quad i=0,\dots,N-1\end{aligned}$$

where $\hat{\boldsymbol{y}}_{N|k}^*:=\hat{\boldsymbol{x}}_{N|k}^*-\hat{\boldsymbol{x}}_{k+N}^r$. Since $\{\hat{\boldsymbol{x}}_{i|k}^*,\hat{\boldsymbol{u}}_{i|k}^*\}$ is feasible at time step k, it follows that, for $i=0,\dots,N-2$,

$$\hat{\boldsymbol{x}}_{i|k+1}^c=\hat{\boldsymbol{x}}_{i+1|k}^*\in\hat{\mathbb{X}},\quad \hat{\boldsymbol{u}}_{i|k+1}^c=\hat{\boldsymbol{u}}_{i+1|k}^*\in\hat{\mathbb{Z}}.$$

For the terminal step, Assumption 1 implies

$$\begin{aligned}\hat{\boldsymbol{u}}_{N-1|k+1}^c&=\kappa_f(\hat{\boldsymbol{y}}_{N|k}^*)\in\hat{\mathbb{Z}}\\ \hat{\boldsymbol{x}}_{N|k+1}^c&=f_{RK}(\hat{\boldsymbol{x}}_{N|k}^*,\kappa_f(\hat{\boldsymbol{y}}_{N|k}^*))\in\mathbb{T}_f.\end{aligned}$$

Therefore, $\{\hat{\boldsymbol{x}}_{i|k+1}^c,\hat{\boldsymbol{u}}_{i|k+1}^c\}$ satisfies

$$\hat{\boldsymbol{x}}_{i|k+1}^c\in\hat{\mathbb{X}},\quad \hat{\boldsymbol{u}}_{i|k+1}^c\in\hat{\mathbb{Z}},\quad \hat{\boldsymbol{x}}_{N|k+1}^c\in\mathbb{T}_f,$$

i.e., all NMPC constraints (state, input, terminal) are fulfilled at time $k+1$ whenever they are fulfilled at time k.

By Assumption 1, we have

$$l_T(\boldsymbol{y}_{N+1|k}) + l(\boldsymbol{y}^*_{N|k}, \kappa_f(\boldsymbol{y}^*_{N|k})) \leq l_T(\boldsymbol{y}^*_{N|k}).$$

Then, the associated cost holds:

$$\begin{aligned} J(\boldsymbol{y}_{k+1}, \hat{\boldsymbol{u}}_{k+1}) &= J^*(\boldsymbol{y}_k, \hat{\boldsymbol{u}}_k) - l(\boldsymbol{y}_{0|k}, \boldsymbol{\mu}_{nmpc}(\hat{\boldsymbol{x}}_k)) \\ &-l_T(\boldsymbol{y}^*_{N|k}) + l(\boldsymbol{y}^*_{N|k}, \kappa_f(\boldsymbol{y}^*_{N|k})) + l_T(\boldsymbol{y}_{N+1|k}) \\ &\leq J^*(\boldsymbol{y}_k, \hat{\boldsymbol{u}}_k) - l(\boldsymbol{y}_{0|k}, \boldsymbol{\mu}_{nmpc}(\hat{\boldsymbol{x}}_k)) \end{aligned} \tag{91}$$

which is an upper bound for $J^*(\boldsymbol{y}_{k+1}, \hat{\boldsymbol{u}}_{k+1})$. Then we have

$$J^*(\boldsymbol{y}_{k+1}, \hat{\boldsymbol{u}}_{k+1}) - J^*(\boldsymbol{y}_k, \hat{\boldsymbol{u}}_k) \leq -l(\boldsymbol{y}_{0|k}, \boldsymbol{\mu}_{nmpc}(\hat{\boldsymbol{x}}_k)). \tag{92}$$

It is evident that the inequality (92) holds for all $\hat{\boldsymbol{x}}_k \in \hat{\mathbb{X}}$, and the candidate control sequence is an admissible solution to (71) at time step $k+1$. Thus, the recursive feasibility of (71) and the stability of (53) are rigorously guaranteed.

F. Proof of Theorem 3

The proof consists of three parts. Part I proposes the asymptotic stability for the original uncertain system, Part II shows the boundedness of internal model states, and Part III demonstrates the robust satisfaction of the constraints.

Part I: Trajectory tracking and disturbance rejection.

By Proposition 1, the NMPC optimization problem (71) is recursively feasible. Moreover, from any feasible time $k \in \mathbb{N}_{\geq 0}$, with the optimal control input $\hat{\boldsymbol{u}}^*_{0|k}$ of (71), the tracking errors $\boldsymbol{y}_k$ asymptotically converge to the origin as $k \to \infty$. By (70), we have $\hat{\boldsymbol{x}}_k \to \hat{\boldsymbol{x}}^r_k$ as $k \to \infty$. By Theorem 2, $\boldsymbol{x}_e$ asymptotically converges to the origin and the nominal system state $\hat{\boldsymbol{x}}$ asymptotically converges to the original system state $\boldsymbol{x}$. Thus, the original system state $\boldsymbol{x}$ asymptotically converges to the time-optimal reference trajectory $\boldsymbol{x}^r$. That is, the agile trajectory tracking and disturbance rejection is achieved.

Part II: Boundedness of internal model states.

Since for $i \in \{f, \tau\}$, M_i is Hurwitz, the subsystem defined by (48e) and (48f) is asymptotically stable, that is, $\lim_{t\to\infty}(\tilde{\boldsymbol{\vartheta}}_f(t), \tilde{\boldsymbol{\vartheta}}_\tau(t)) = (\boldsymbol{0}, \boldsymbol{0})$. Thus, there exists a compact set $\bar{\mathbb{V}}_i$ containing the origin such that $\tilde{\boldsymbol{\vartheta}}_i(t) \in \bar{\mathbb{V}}_i$. By applying transformation (30), (33), (38), (42) and (45), we obtain

$$\begin{aligned} \boldsymbol{\vartheta}_f &= m^{-1}\boldsymbol{\theta}_f + N_f \boldsymbol{v}_e + \tilde{\boldsymbol{\vartheta}}_f \\ \boldsymbol{\vartheta}_\tau &= \boldsymbol{\theta}_\tau + N_\tau J \boldsymbol{\Omega}_e + \boldsymbol{z}\boldsymbol{\gamma} + \tilde{\boldsymbol{\vartheta}}_\tau. \end{aligned}$$

By Theorem 2 and Proposition 1, we have $\lim_{t\to\infty}(\boldsymbol{v}_e(t), \boldsymbol{\Omega}_e(t), \boldsymbol{q}_{ev}(t), \boldsymbol{z}(t)) = (\boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0})$. Thus, the internal model states $\boldsymbol{\vartheta}_f$ and $\boldsymbol{\vartheta}_\tau$ will asymptotically converge to $m^{-1}\boldsymbol{\theta}_f$ and $\boldsymbol{\theta}_\tau$ as $t \to \infty$, respectively. Thus, there always exist compact sets $\mathbb{V}_f$ and $\mathbb{V}_\tau$ such that $\boldsymbol{\vartheta}_f \in \mathbb{V}_f$ and $\boldsymbol{\vartheta}_\tau \in \mathbb{V}_\tau$.

Part III: Robust constraint satisfaction.

Recalling that (71) is recursively feasible, we have $\hat{\boldsymbol{x}}_k \in \hat{\mathbb{X}}$ and $\hat{\boldsymbol{u}}_k \in \hat{\mathbb{Z}}$ for all $k \in \mathbb{N}_{\geq 0}$. By Theorem 2 and (64), the states of the original system (1)-(4) satisfy $\boldsymbol{x}_k = \hat{\boldsymbol{x}}_k + \boldsymbol{x}_{e,k} \in (\hat{\mathbb{X}} \oplus \mathbb{V}_0) \subseteq \mathbb{X}$. Also, $\bar{\boldsymbol{u}}_{i,k} = \hat{\boldsymbol{u}}^*_{0|k} + \boldsymbol{\delta}_{i,k} \in \hat{\mathbb{Z}}_i \oplus \Delta_i \subseteq \mathbb{Z}_i$, where $\mathbb{Z}_i := \bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1} \mathbb{V}_i$.

Let $a \in \{\bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1}\mathbb{V}_i\} \oplus \Psi_i T_i^{-1}\mathbb{V}_i$. By Minkowski set addition, we have $a := \{x + y | x \in \bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1}\mathbb{V}_i, y \in \Psi_i T_i^{-1}\mathbb{V}_i\}$. Since $\bar{\boldsymbol{u}}_i \in \bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1}\mathbb{V}_i$, it follows that $\bar{\boldsymbol{u}}_i + b \in \bar{\mathbb{U}}_i, \forall b \in \Psi_i T_i^{-1}\mathbb{V}_i$. Specifically, if $b = y$, we have $\bar{\boldsymbol{u}}_i + y \in \bar{\mathbb{U}}_i$. Thus, for any $a \in \{\bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1}\mathbb{V}_i\} \oplus \Psi_i T_i^{-1}\mathbb{V}_i$, it follows that $a \in \bar{\mathbb{U}}_i$. From Part II, the internal model states satisfy $\boldsymbol{\vartheta}_i \in \mathbb{V}_i$, implying that $\Psi_i T_i^{-1}\boldsymbol{\vartheta}_i \in \Psi_i T_i^{-1}\mathbb{V}_i$. Thus, by transformation (30) and (38), we have

$$\begin{aligned} \boldsymbol{u}_{i,k} &= \bar{\boldsymbol{u}}_{i,k} + \Psi_i T_i^{-1}\boldsymbol{\vartheta}_{i,k} \\ &\in (\{\bar{\mathbb{U}}_i \ominus \Psi_i T_i^{-1}\mathbb{V}_i\} \oplus \Psi_i T_i^{-1}\mathbb{V}_i) \subseteq \bar{\mathbb{U}}_i, i \in \{f, \tau\}. \end{aligned}$$

By (29), the original control inputs (6) satisfy

$$f_T \in \Pi_{\boldsymbol{z}_B}\left(\bar{\mathbb{U}}_f \oplus \mathbb{D}_r^{\boldsymbol{v}^r} \ominus \mathbb{G}_f\right) \subseteq \mathbb{U}_f, \quad \boldsymbol{\tau} \in \mathbb{U}_\tau$$

where the projection map $\Pi_{\boldsymbol{z}_B}(\mathcal{S}) := \{\boldsymbol{s} \cdot \boldsymbol{z}_B | \boldsymbol{s} \in \mathcal{S}\}$. Therefore, the constraints on the states and control inputs in Problem 2 are always satisfied. The proof is thus complete.

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